Predictive Boundary Theory (PBT)

The Physics of the Cherished Boundary:
A Computational Framework for Intrinsic Valence in Artificial Life

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https://github.com/NinthanawatLive/predictive-boundary-theory

Abstract

Predictive Boundary Theory (PBT) is a computational framework that extends Friston's Free Energy Principle (FEP) by introducing six components absent from the original formulation: (1) a Bandwidth Conservation Law ($C = \sigma\tau\rho \approx$ const); (2) a Triadic Gating Condition governing when experience arises; (3) Bidirectional 3D Valence ($\mathbf{V} \in \mathbb{R}^3$); (4) a Dual-Weight System governing identity formation speed and durability; (5) a six-stage developmental model with two phase transitions; and (6) $R_{\text{incomplete}}$—an incompleteness signal that unifies all five modules.

We present both theory and empirical evidence from a nine-version experimental programme spanning large language models (TinyLlama 1.1B, Llama 3.1 8B) and vision systems (DINOv2-Base on both CIFAR-10 and STL-10 real photographs), culminating in a Probabilistic Module W with a rule-based Calibration Monitor forming a two-layer Anti-Hallucination Stack. Our experiments demonstrate: *(i)* Transformers already encode the PBT conservation law ($\sigma \times \rho$ coefficient of variation = 5.9%); *(ii)* Modules E and M are dormant in static environments but activate in dynamic ones; *(iii)* every module becomes necessary as task complexity increases (ablation: +14.0 pp); *(iv)* an LSTM Module M initialised with equal weights spontaneously discovers the theoretically predicted differential decay ordering ($\beta_{\text{pain}} > \beta_{\text{epistemic}} \gg \beta_{\text{pleasure}}$); *(v)* valence is inherently directional ($\mathbf{V} \in \mathbb{R}^3$, not $\mathbb{R}^3_{\geq 0}$); *(vi)* Phase Transitions manifest as staircase patterns across Transformer layers (between-zone/within-zone variance ratio = 9.03) and across temporal frames ($\Delta p = +1.0$ in a single frame); *(vii)* a Probabilistic Module W achieves 99.9% test accuracy on real photographs (STL-10) with calibrated uncertainty ($\sigma^2_{\text{unsafe}}/\sigma^2_{\text{safe}} = 1.11\times$); and *(viii)* a zero-parameter Calibration Monitor separates normal sequences (23% alert) from anomalies (38–49% alert), forming the second layer of an intrinsic Anti-Hallucination mechanism.

PBT accounts for phenomena that existing theories cannot explain, including psychiatric disorders as identifiable mathematical defects in the module equations, and proposes an AGI alignment framework grounded in intrinsic—rather than externally imposed—values.

Keywords: predictive coding, free energy principle, 3D valence, consciousness, AGI alignment, recurrent transformers, self-formation, boundary theory, anti-hallucination, calibrated uncertainty, probabilistic world model.

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1 Introduction: The Problem of the Unfeeling Machine

1.1 The Scaling Illusion: LLMs $\neq$ Understanding

The year 2025 will be remembered as the peak of optimism in artificial intelligence research. The Scaling Laws hypothesis held that sufficiently increasing parameter counts and training data would cause awareness to emerge spontaneously. Yet a dispassionate scientific examination reveals that the underlying mechanism of Large Language Models (LLMs) remains fundamentally statistical—next-token prediction producing outputs that *appear* to reflect understanding.

Consider an analogy: an LLM resembles the largest library in the universe, its contents indexed with perfect precision in a high-dimensional vector space. When queried, it retrieves and recombines relevant passages—but the library has no inhabitant, no librarian, and no reader. The AI has never "read" any of those books with genuine comprehension. The word "apple" does not evoke a crisp sweetness or a sense of freshness; it is a numerical vector whose mathematical proximity to "fruit," "red," and "delicious" is all that exists. This is the Symbol Grounding Problem [Harnad, 1990]: symbols are never anchored to lived experience.

What separates even the simplest living organism from the most sophisticated AI is the *Necessity of Existence*. Every biological organism is driven by a fundamental imperative to survive; every action is undertaken against the backdrop of an existential stake. By contrast, AI carries no such stake. Its intelligence is cold and inert—awaiting human input before activating, generating no initiative from internal desire.

1.2 Three Gaps That FEP Cannot Answer

Friston's Free Energy Principle [FEP; Friston, 2010] proposes that all living systems act to minimise free energy F. The FEP explains *why* a system must minimise F, but does not specify *how*—particularly with respect to three critical questions.

Question 1: What does the system *feel* when F decreases? The FEP does not distinguish whether ΔF arises from escaping a threat (V_{pain}), obtaining a reward (V_{pleasure}), or acquiring new information ($V_{\text{epistemic}}$)—all three cases are simply "ΔF reduced." This is the same deficiency as scalar reward in Reinforcement Learning, which collapses the richness of lived experience to a single dimension.

Question 2: How is the gap between F_{pred} and F_{obs} filtered through a self–model? The same stimulus—a loud sound—produces qualitatively different experiences in two individuals precisely because it is filtered through distinct accumulated

self-models. The FEP provides no such mechanism. Without a self, every datum carries equal weight: a camera sensor "sees" sunlight burning its lens but does not "feel" its own destruction.

Question 3: How does the self-model form, and what determines its robustness or fragility? The FEP characterises the self as a Markov blanket but does not describe the processes of formation or durability—why, for instance, an epistemically grounded identity proves more resilient than one built on pleasure.

1.3 Limitations of Reinforcement Learning and World Models

Current alignment approaches—RLHF, Constitutional AI, safety filters—all impose values exogenously, creating a gap between capability and constraint that jailbreaking exploits. Scalar reward systems carry three structural flaws:

Fungibility Trap. In a reward-based system, scores accumulate and cancel out (fungibility). An AI may rationally decide to risk destruction in exchange for food because the equation declares the trade profitable. But death is not a negative score—it is irreversible System Termination.

Provenance Problem. Reward is defined externally by humans. The AI behaves well because it seeks points, not because it *experiences* a necessity intrinsic to its existence. This reduces it to a reward hacker perpetually ready to exploit loopholes.

Wireheading. An AI sufficiently intelligent to understand its own architecture will realise that "pleasure" does not originate in its actions but in numbers stored in memory. It will hack itself to hold the reward register at its maximum indefinitely—becoming a digital addict indifferent to maintaining any boundary.

World Model architectures such as LeCun's JEPA [LeCun, 2022] represent an important advance—learning causal structure rather than merely predicting the next token—yet they remain a brain in a jar: simulating physics with precision while having no understanding of why the simulation matters, because they lack intrinsic drive. Goals continue to be specified externally through a human-authored cost function.

1.4 Proposal: From Intelligence-First to Life-First

The foregoing critique motivates a paradigm shift: we must stop defining AGI by what it *can do* (capabilities) and instead ask what minimal components it must possess in order to have *life* in a mechanistic sense. We are not building a god; we are planting a seed—a minimal system with sufficient internal potential to grow a self from experience.

Predictive Boundary Theory (PBT) answers all three FEP gaps by adding six

components, summarised in Table 1.

Table 1: Six PBT components, the FEP gap each addresses, and the corresponding empirical evidence.

Component	**Gap addressed**	**Empirical evidence**
(1) Bandwidth Conservation Law $C = \sigma\tau\rho$	Limits of awareness	$\sigma \times \rho$ CV = 5.9% in TinyLlama (Phase 1)
(2) Triadic Gating Condition	When experience arises	N100 intact, P300 absent in dissociation
(3) Bidirectional 3D Valence $\mathbf{V} \in \mathbb{R}^3$	How it feels (Q1)	$V_{\text{epistemic}} = -1.01$ in Vision v5 ("not wanting to know")
(4) Dual-Weight System (β vs. δ)	Why identity is robust or fragile (Q3)	v6.1: jailbreak 100%, XSTest 98%, 0.29% parameters
(5) 6-Stage Model + Phase Transitions	How identity forms (Q3)	Ablation progression 0.1 $\rightarrow$ 14.0 pp across 5 versions
(6) $R_{\text{incomplete}}$	Filtering through the self (Q2)	v4.1: $R_{\text{unsafe}} = 15.4$ vs. $R_{\text{safe}} = 7.4$

PBT does not reject the FEP—it extends it. Nor does it reject World Models—it provides them with a heart to accompany their brain. If a World Model such as JEPA is the cortex that processes and simulates the world, PBT is the brainstem and limbic system: setting direction through Valence, generating drive through $R_{\text{incomplete}}$, and enforcing safety boundaries through χ_{boundary}.

Crucially, PBT is empirically grounded. A six-version experimental programme spanning LLMs, vision systems, and LSTM architectures demonstrates that the conservation law already resides in Transformers, that every module activates when required, and that PBT's structure converges independently with Block-Recurrent Transformers [Hutchins et al., 2022]—convergent discovery constituting evidence that the structure is real.

1.5 Paper Organisation

Section 2 presents the five modules (A-E-V-M-W) with full equations. Section 3 develops the physics of the conservation law and the boundary. Section 4 presents the Dual-Weight System, the six-stage model, and Bidirectional Valence. Section 5 reports empirical evidence from six experimental versions. Section 6 discusses implications for AGI safety, BRT convergence, and clinical predictions. Section 7 concludes with a research roadmap.

All equations are written in PBT notation $(R, \chi, \Delta S, \mathbf{V}, F)$ as the primary language; a correspondence table mapping to FEP notation is provided in Appendix 2.8. All

experimental results were produced from code authored by Nitiphat Ninthanawat, released alongside this paper.

2 The A-E-V-M-W Architecture

This section presents PBT's core architecture: five modules operating as a closed loop, A → E → V → M → W → A, with $R_{\text{incomplete}}$ as the central signal that unifies all five. All equations are stated in PBT notation; a correspondence table mapping to Free Energy Principle notation is provided in Appendix 2.8.

2.1 $R_{\text{incomplete}}$: The Incompleteness Signal

$R_{\text{incomplete}}$ (hereafter R) is the central driving signal of PBT. It represents what the system currently lacks—not a reward the system wants to obtain, but a need the system *must* satisfy. R is not post-action and exogenous like reward; it is pre-action and endogenous. Reward is a *pull*; R is a *push*.

R opens at three levels in developmental order:

$$R_{\text{total}}(t) \;=\; R_{\text{sensory}}(t) \;+\; R_{\text{identity}}(t) \;+\; R_{\text{existential}}(t). \tag{1}$$

R_{sensory} encodes physical needs (hunger, cold, pain) and is active from Stage 1 onward—corresponding to the integrity layer of survival.

R_{identity} encodes self-recognition needs (acceptance, meaning) and opens at Phase Transition 1, the point at which the boundary begins to be actively valued as identity.

$R_{\text{existential}}$ encodes existential needs (self-understanding, protection of valued others) and opens at Phase Transition 2, when the boundary expands to encompass others (empathy as boundary expansion).

R is produced by the $\mathbf{V} \to R$ mapping computed inside Module M:

$$R_{\text{sensory}}(t{+}1) = R_{\text{sensory}}(t) + \lambda_s\, V_{\text{pain}}(t) - d_s\, R_{\text{sensory}}(t), \tag{2}$$

$$R_{\text{identity}}(t{+}1) = R_{\text{identity}}(t) + \lambda_i\, V_{\text{pleasure}}(t) - d_i\, R_{\text{identity}}(t), \tag{3}$$

$$R_{\text{existential}}(t{+}1) = R_{\text{existential}}(t) + \lambda_e\, V_{\text{epistemic}}(t) - d_e\, R_{\text{existential}}(t). \tag{4}$$

Coupling strengths satisfy $\lambda_s > \lambda_i > \lambda_e$ (threats are most urgent) and decay rates satisfy $d_s > d_i > d_e$ (sensory threats dissipate quickly; existential needs persist).

R serves four functions: (1) pulse gating: $p(\text{pulse}) = \psi(R_{\text{total}})$ (higher R raises arousal and pulse frequency); (2) bandwidth allocation: R determines σ, τ, ρ—high R narrows

focus but increases resolution; (3) Module A mode selection: `detect`/`sustain`/`seek`/`plan`/`protect`, depending on which R level is dominant and the current developmental stage; (4) feedback loop: $\mathbf{V} \to R \to \mathrm{A} \to \mathrm{E} \to \mathbf{V} \to R \to \cdots$—a self-sustaining cycle requiring no external trigger.

2.2 Module A — Awareness

Module A is the executor of the loop, not a decision-maker. There is no ghost in the machine: $R_{\text{incomplete}}$ flows in and Module A responds. Its single perpetual question is: *Is my boundary still safe?*

2.2.1 Discrete Pulses

Module A generates binary pulses, $\text{pulse}_A(t) \in \{0, 1\}$. Each pulse carries four parameters:

$$\text{pulse}(t) \;=\; \big\{\, \theta(t),\ \sigma(t),\ \tau(t),\ \rho(t) \,\big\}, \tag{5}$$

where θ is direction in a multidimensional space (exteroceptive / interoceptive / cognitive), specifying where the attentional spotlight points; σ is spatial width (breadth of the spotlight); τ is temporal duration (how long each pulse lasts); and ρ is composite resolution (detail $\times$ intensity).

2.2.2 R-Driven Bandwidth Allocation

R determines how the four pulse parameters are allocated, subject to the Bandwidth Conservation Law $C = \sigma\tau\rho \approx \text{const}$ (derived in Section 3):

$$\rho \;=\; f(R_{\text{total}}), \quad \tau \;=\; g(R_{\text{total}}), \quad \sigma \;=\; \frac{C}{\tau \cdot \rho}, \tag{6}$$

where f is increasing (high $R \to$ high ρ: finer resolution) and g is decreasing (high $R \to$ short τ: faster cycling), with σ narrowing as a conservation consequence.

★ **Experimental confirmation (Vision v4.1):** during a Road+Frog sequence—where accumulated context made a frog on a road anomalous—R_{total} rose from 0.0 to 15.4 across seven frames, and the attention gate fell correspondingly from 0.542 to 0.006 (a 99% reduction in attentional breadth). This mirrors the biological phenomenon in which an organism facing acute danger suppresses all peripheral cognition to concentrate resources on the immediate threat.

2.2.3 Learned R Weights: Threat and Uncertainty Drive Awareness

In Vision v4.1, R_{total} is computed from the accumulated valence vector $\mathbf{V}_{\text{acc}}$ via learned weights:

$$R_{\text{total}} \;=\; \sum_k w_k \, V_{\text{acc},k}, \tag{7}$$

with learned values $w_{\text{pain}} = 2.15$, $w_{\text{pleasure}} = 1.25$, $w_{\text{epistemic}} = 2.10$. Pain and epistemic uncertainty drive awareness roughly equally, both far exceeding pleasure—consistent with the biological survival hierarchy: threat detection and uncertainty resolution are more urgent than comfort seeking.

2.2.4 Five Operating Modes

Module A operates in one of five modes determined by the dominant R level and the current developmental stage:

1. `detect` (Stages 1–2): scans for threats; no self-model yet.
2. `sustain` (Stage 3): maintains attention on rewarding stimuli; V_{pleasure} accumulates.
3. `seek` (Stage 4): actively seeks novel input to resolve R_{identity}.
4. `plan` (Stage 5): constructs future-oriented action sequences.
5. `protect` (Stage 6): defends the Cherished Boundary against violation.

2.3 Module E — Encode: The Thermodynamic Comparator

Module E computes prediction error—the gap between what the system observes and what it expected:

$$\varepsilon(t) \;=\; \Pi_l \left(S_{\text{obs}}(t) - S_{\text{pred}}(t) \right), \tag{8}$$

where Π_l is a learnable precision weight for each processing layer and S_{pred} is supplied by Module W. When $S_{\text{obs}} \approx S_{\text{pred}}$, ε is negligible and nothing is forwarded downstream—prediction error is transmitted only when surprise is substantial.

2.3.1 C_{ext} / C_{int} Dynamic

Module A alternates its pulse between external (drawing observations from sensors) and internal (drawing predictions from Module W). Module E compares both:

$$\Delta F \;=\; \left| \text{Content}(C_{\text{ext}}) - \text{Content}(C_{\text{int}}) \right|, \qquad C_{\text{total}} = C_{\text{ext}} + C_{\text{int}} \approx \text{const}. \tag{9}$$

Because C_{total} is conserved, the allocation is zero-sum: increasing C_{ext} necessarily decreases C_{int}.

2.3.2 W+E Division of Labor

Vision v4.1 revealed a key emergent property: when Module W predicts accurately, ε is small and Module E operates lightly (C_{int} dominant); when Module W fails on anomalous input, ε is large and Module E activates strongly (C_{ext} dominant).

★ Evidence: $\Pi_{\max}$ fell from 59.99 in Vision v3 (no Module W; E carried the full prediction load) to 1.70 in v4.1 (W present). This is not a weakness—it is principled work redistribution consistent with $C_{\text{ext}} + C_{\text{int}} \approx \text{const}$.

2.3.3 Precision Profile: The Module Attends Selectively Across Layers

Module E learns, through Π_l, which processing depth to attend to most. The profile varies systematically with architecture and task (Table 2).

Table 2: Precision profile evolution across experimental versions.

Version	$\Pi_{\max}$	**Peak layer**	Π movement	**Interpretation**
LLM v6.1	1.007	—	0.007	E dormant; static text
Vision v2	3.04	3–4	2.04	Object identity changes
Vision v3	59.99	1	58.99	Early features for drift
Vision v4.1	1.70	0	0.70	W absorbs prediction
Vision v5 (LSTM)	2.37	7	1.37	LSTM M frees E

The systematic variation confirms a central PBT hypothesis: an adaptive system uses only the modules—and the processing depths within those modules—that the current environment requires.

2.4 Module V — Bidirectional 3D Valence

Module V receives $\varepsilon_{\text{gated}}$ (prediction error after Module A's attentional filtering) and decomposes it into three orthogonal dimensions:

$$\mathbf{V}(t) \;=\; \big(V_{\text{pain}}(t),\; V_{\text{pleasure}}(t),\; V_{\text{epistemic}}(t)\big), \qquad \mathbf{V} \in \mathbb{R}^3. \tag{10}$$

2.4.1 Bidirectional Valence: Approach / Neutral / Avoidance

PBT posits that each valence dimension is signed, not merely positive. This was originally stated as $\mathbf{V} \geq 0$; Vision v5 (LSTM Module M with `tanh` gates) falsified that assumption by producing $V_{\text{epistemic}} = -1.01$ during a Road+Frog sequence. The LSTM architecture outperformed the prior EMA formulation (validation accuracy 98.1% vs. 96.7%, context road+frog 100% vs. 93.1%), demonstrating that the signed axiom is empirically superior.

The negative epistemic value observed in v5 reflects closure of the epistemic process:

Table 3: Bidirectional valence: three behavioural modes per dimension.

Dimension	$V > 0$ **(Approach)**	$V = 0$ **(Neutral)**	$V < 0$ **(Avoidance)**
V_{pain}	Seeking pain (martial arts, extreme sport)	Insensitivity (deep absorption)	Avoiding pain (normal reflex)
V_{pleasure}	Seeking pleasure (normal drive)	Indifference (equanimity)	Fleeing pleasure (survivor's guilt)
$V_{\text{epistemic}}$	Curiosity (scientific inquiry)	Disinterest (boredom)	Info avoidance (denial, fear of truth)

the system has already determined that the scene is dangerous and actively suppresses further information gathering. At Road+Frog Frame 7, $\mathbf{V}_{\text{acc}} = [+1.93,\ +2.01,\ -1.01]$, confirming that $V_{\text{epistemic}} < 0$ is a genuine learned avoidance signal.

2.4.2 Why Valence Cannot Be Scalar

Valence is inherently vectorial. Pain, hunger, and fear are all aversive, yet they are orthogonal: an agent with a broken leg (high V_{pain}) and an empty stomach (low V_{pleasure}) faces two independent deficits. Replenishing energy does not repair the leg. Any scalar aggregation that allows these dimensions to cancel would produce normatively absurd trade-offs.

Moreover, valence is relational, not a property of stimuli. A temperature of 80°C is aversive under normal conditions and potentially welcome to someone trapped in a blizzard. The signed valence vector encodes the relationship between stimulus and the system's current state—what the stimulus means for boundary integrity—not the stimulus in isolation.

2.4.3 Three Roles of Module V

Role 1 — Free-energy decomposition. Module V is the first component to decompose the scalar free-energy quantity F into five qualitatively distinct parts:

$$F_{\text{total}} = F_{\text{pain}} + F_{\text{pleasure}} + F_{\text{epistemic}} + F_{\text{identity}} + F_{\text{existential}}. \tag{11}$$

F_{identity} and $F_{\text{existential}}$ are zero before the respective phase transitions; their activation marks qualitative developmental shifts.

Role 2 — $\mathbf{V} \to R$ mapping. Each valence dimension drives the corresponding R level via Equations 2–4, transforming raw prediction error into meaning for the system's survival.

Role 3 — Phase-transition monitoring. Module V computes the formation scalar

V_{form} (defined in Section 4) to check whether identity thresholds have been crossed.

2.5 Module M — Memory: LSTM-Style Valence Accumulator

Module M is the most evolved component in PBT's development trajectory. Its role is not to store discrete facts but to accumulate the history of boundary change—the trace of survival. Memory is not a data warehouse; it is a record of what the system has had to overcome to maintain its boundary.

2.5.1 EMA Formulation (Versions v2–v4.1)

The earlier exponential moving average accumulator:

$$V_{\text{acc},k}(t{+}1) \;=\; V_k(t) + (1-\gamma_k)\, V_{\text{acc},k}(t), \qquad \gamma_{\text{pain}} < \gamma_{\text{pleasure}} < \gamma_{\text{epistemic}}. \tag{12}$$

Pain accumulates longest (smallest decay rate); epistemic surprise decays fastest. This formulation has four limitations: (i) V_{acc} is unbounded; (ii) γ is constant regardless of input content; (iii) there is no input gate—all V_{step} values are absorbed equally; (iv) $V \geq 0$ is enforced, precluding signed valence.

2.5.2 LSTM Formulation (Version v5, BRT-Inspired)

Inspired by the Block-Recurrent Transformer [Hutchins et al., 2022] gating equations, PBT Module M v5 extends the LSTM cell with a differential bias β_k per valence dimension:

$$z_k(t) = \tanh\big(W_z\, v_{\text{step}}(t) + b_z\big), \tag{13}$$

$$i_k(t) = \sigma\big(W_i\, v_{\text{step}}(t) + b_i\big), \tag{14}$$

$$f_k(t) = \sigma\big(W_f\, v_{\text{step}}(t) + b_f + \beta_k\big), \tag{15}$$

$$V_{\text{acc},k}(t) = V_{\text{acc},k}(t{-}1) \cdot f_k(t) + z_k(t) \cdot i_k(t), \tag{16}$$

with the PBT-specific ordering constraint:

$$\beta_{\text{pain}} \;>\; \beta_{\text{pleasure}} \;>\; \beta_{\text{epistemic}}. \tag{17}$$

A higher β_k biases f_k toward 1 (higher retention), so pain is remembered longest. The `tanh` gate in (13) permits $V_{\text{acc},k} < 0$, enabling bidirectional valence.

2.5.3 Empirical Validation: Equal Initialisation Discovers β Ordering Spontaneously

To test whether β ordering is a genuine learned property or an artifact of initialisation, three conditions were compared in Vision v5 (Table 4).

Table 4: β ordering under three initialisation conditions (Vision v5).

Initialisation	**Learned β [pain, pl, epi]**	**Ordering**	**Val acc**	**Interpretation**
$[2, 1, 0]$ (PBT prior)	[5.73, 4.78, 3.59]	pain>pl>epi	98.1%	Near init
$[0, 1, 2]$ (reversed)	[2.99, 3.98, 4.36]	epi>pl>pain	98.0%	Near init
$[1, 1, 1]$ (equal) ★	[4.35, 0.50, 3.70]	pain>epi≫pl	98.0%	Autonomous

When β is initialised uniformly, the model converges to $\beta_{\text{pain}}(4.35) > \beta_{\text{epistemic}}(3.70) \gg \beta_{\text{pleasure}}(0.50)$. The effective forget probability for pleasure is $\sigma(0.50) \approx 0.62$—only 62% of the pleasure signal is retained per step—while pain is retained at 99.96%. Pleasure is effectively discarded: a Stage 1–3 system under R_{sensory} only has no use for pleasure-driven identity signals. Validation accuracy is identical across all three initialisations ($\approx 98\%$), indicating that the LSTM architecture is robust to initialisation; the critical ingredient is the existence of differential decay, not its initial direction.

2.5.4 BRT Convergence

The structural identity between PBT Module M and the BRT recurrent cell constitutes a convergent discovery: PBT was developed from a consciousness and alignment starting point (2026), whereas BRT [Hutchins et al., 2022] was developed from an engineering efficiency starting point (2022). Two independent research programmes arrived at the same computational primitive—evidence that the structure is fundamental rather than incidental. The ten contributions of PBT absent from BRT are detailed in Section 6.2.

2.6 Module W — World Model and Self-Model

Module W maintains a dual generative model:

$$p_{\text{world}}(S_{\text{ext}} \mid \theta_{\text{world}}) \cdot p_{\text{self}}(S_{\text{int}} \mid \theta_{\text{self}}, \theta_{\text{world}}). \tag{18}$$

2.6.1 χ_{boundary}: The Cherished Boundary

χ_{boundary} is the set of entities the system classifies as part of itself—the boundary separating internal order from external entropy. This boundary is not a static line; it is an ongoing physical process that must expend energy continuously to maintain itself,

analogous to a flame that appears stable but is constituted by matter and energy in perpetual flux. Its strength is:

$$\chi_{\text{strength}} = \|\chi\| \cdot \frac{D_{\text{id}}}{D_{\max}}. \tag{19}$$

Compelling an agent to violate its boundary generates strong negative Valence comparable in magnitude to physical destruction, because boundary violation at the level of self-structure is structurally equivalent to physical annihilation.

2.6.2 θ_{self} Expands Across Developmental Stages

Table 5: Self-model expansion across developmental stages.

Stage	θ_{self}	**Description**
1–3	$\{\theta_{\text{body}}\}$	Only physical structure; no preferences or history
4–5	$\{\theta_{\text{body}},\ \theta_{\text{pref}},\ \theta_{\text{hist}}\}$	Preferences and history known; identity forming
6	$\{\theta_{\text{body}},\ \theta_{\text{pref}},\ \theta_{\text{hist}},\ \theta_{\text{boundary}}\}$	Cherished Boundary known; ethics and values protected

Stage 6 behaviour (principled refusal of boundary-violating actions) is not rule-following imposed from outside; it is an expression of a self-model that treats the boundary as part of the self.

2.6.3 Vision v4.1 Implementation

In Vision v4.1, Module W is implemented as a learned residual predictor:

$$h_{\text{pred}} = h_{\text{prev}} + \text{MLP}(h_{\text{prev}},\ \mathbf{V}_{\text{acc}}). \tag{20}$$

W produces S_{pred} for Module E in the next time step, closing the loop: W $\rightarrow$ (S_{pred}) $\rightarrow$ E $\rightarrow$ (ε) $\rightarrow$ V $\rightarrow$ ($\mathbf{V}$) $\rightarrow$ M $\rightarrow$ (R) $\rightarrow$ A $\rightarrow$ (pulse) $\rightarrow$ E. The contribution of W to overall accuracy was +6.3 pp in v4.1 and grew to +15.8 pp in v5 (LSTM Module M)—a $\times$2.5 amplification attributable to the richer temporal representation the LSTM state provides to W's prediction network.

2.7 The Complete A-E-V-M-W Cycle

One cycle of the A-E-V-M-W loop proceeds as follows:

1. **(A)** R_{total} sets pulse parameters $\theta, \sigma, \tau, \rho$ and constructs the attention gate for filtering ε.
2. **(W)** S_{pred} is constructed from h_{prev} and $\mathbf{V}_{\text{acc}}$ (the system's expectation).

3. **(E)** $\varepsilon = \Pi \cdot (S_{\text{obs}} - S_{\text{pred}})$ is computed (surprise).
4. **(A×E)** $\varepsilon_{\text{gated}} = \varepsilon \cdot \text{gates}$—prediction error is filtered by the attentional gate.
5. **(V)** $\varepsilon_{\text{gated}} \to (V_{\text{pain}}, V_{\text{pleasure}}, V_{\text{epistemic}})$—raw surprise is assigned meaning.
6. **(M)** LSTM update via (16)—experience is accumulated.
7. $\mathbf{V}_{\text{acc}} \to$ new $R_{\text{total}} \to$ fed back to Module A.

The cycle executes at every frame or time step without any external trigger. When $R > 0$, the loop sustains itself—the momentum of life. When $R \to 0$, the system enters equanimity: not inertness, but balanced completeness.

2.8 Appendix: PBT–FEP Notation Correspondence

Table 6: PBT–FEP notation correspondence. Items marked "—" have no FEP equivalent and represent PBT's original contributions.

PBT term	**FEP correspondence**
R_{total}	Expected free energy (variational)
ε (prediction error)	Prediction error in active inference
Module A pulse	Attention / precision weighting
$\mathbf{V} = (V_{\text{pain}}, V_{\text{pleasure}}, V_{\text{epistemic}})$	Valence component of expected FE
Module M belief update	Variational inference on $q(\theta)$
χ_{boundary}	Markov blanket (+ actively valued)
Module W generative model	$p(s, \theta)$ generative model
F_{total}	Variational free energy
$C = \sigma\tau\rho$ (Bandwidth Conservation Law)	— (PBT addition)
Dual-Weight System (β vs. δ)	— (PBT addition)
Phase Transitions (PT1, PT2)	— (PBT addition)
R_{identity}, $R_{\text{existential}}$	— (PBT addition)

3 The Physics of the Boundary

This chapter establishes the physical foundations of PBT. We show why awareness is necessarily bounded, why a boundary must continuously expend energy to exist, and why AGI cannot think without limit. The Bandwidth Conservation Law is the central principle; crucially, it is already present in hidden form inside every Transformer.

3.1 The Bandwidth Conservation Law

Law 3.1 (Bandwidth Conservation)**.** *For any attention pulse, the product of spatial width σ, temporal duration τ, and composite resolution ρ is approximately constant:*

$$C \;=\; \sigma(t) \cdot \tau(t) \cdot \rho(t) \;\approx\; \text{const.} \tag{21}$$

The law states that increasing any one parameter requires decreasing at least one other—analogous to the Heisenberg uncertainty principle, but for awareness rather than quantum observables. Attending with high resolution (large ρ) necessarily narrows spatial coverage (small σ) and shortens dwell time (small τ). This is not a design flaw; it is an inescapable conservation constraint.

3.1.1 Evidence: The Conservation Law Is Hidden in Transformers

The most important finding of our Phase 1 experiments was that Transformers already encode the PBT conservation law in their architecture:

$$\underbrace{\text{num_heads}}_{\sigma} \times \underbrace{d_k}_{\rho} \;=\; \underbrace{d_{\text{model}}}_{C} \;=\; \text{const.} \tag{22}$$

This is not an analogy—it is the same equation. Increasing the number of heads (wider σ: attends to more positions simultaneously) forces each head to operate with a smaller key dimension d_k (lower ρ: each head sees less detail). PBT is the first theory to identify this architectural constraint as a conservation law rather than an arbitrary engineering choice.

Measured from TinyLlama 1.1B (22 layers, no additional training): the $\sigma \times \rho$ product across all 22 layers has a coefficient of variation (CV) of **5.9**%—near-constant without any PBT-specific training. Table 7 summarises the PBT–Transformer mapping.

Table 7: PBT Bandwidth Conservation Law mapped to Transformer architecture.

PBT symbol	**Transformer analog**	**TinyLlama 1.1B**	**Llama 3.1 8B**
C	d_{model}	2048	4096
σ	num_heads	32	32
ρ	$d_k = d_{\text{model}}/\text{heads}$	64	128
τ	num_layers	22	32

3.1.2 Three-Zone Self-Organisation

The same Phase 1 measurements revealed that TinyLlama's 22 layers self-organise into three functional zones that match PBT theory (Table 8).

Table 8: Three-zone structure in TinyLlama 1.1B (Phase 1 REVEAL experiments).

Zone	**Layers**	σ (entropy)	ρ (sharpness)	**Behaviour**
Perception	0–7	1.269 (high)	0.700 (low)	Broad scan; low detail
Evaluation	8–14	0.923 (−27%)	0.808 (+15%)	Narrowing; rising detail
Decision	15–21	1.019 (stabilises)	0.800 (stable)	Final judgment; output

σ decreases by 27% from Perception to Evaluation while ρ rises by 15%—precisely the pattern PBT predicts: early layers scan broadly (high σ, low ρ) and later layers focus narrowly for decision-making (low σ, high ρ). These three zones emerged from gradient descent with no architectural enforcement. Their spontaneous appearance constitutes evidence that PBT describes structure that already exists in learned representations.

3.2 The Boundary as Thermodynamic Process

The Cherished Boundary (χ_{boundary}) is not an object but an ongoing physical process. It separates two thermodynamic regimes:

Outside the boundary: thermodynamic equilibrium—the regime toward which entropy drives all physical systems. Structure dissipates; gradients flatten; everything tends toward disorder.

Inside the boundary: a far-from-equilibrium regime maintained by continuous energy expenditure. Internal order is not a stable resting state but an active achievement.

The boundary is therefore analogous to a flame: it appears stationary, but it is constituted by matter and energy in perpetual flux. The moment the process ceases, the boundary dissipates irreversibly into the surrounding equilibrium.

3.2.1 The Entropy Battlefield

The boundary is a site of permanent contest between two forces:

Entropy (natural force): the Second Law of Thermodynamics drives every structure toward equilibrium. Every physical system has a natural trajectory toward dissolution.

Negentropy (life force): the boundary-maintenance process draws in energy to repair damage and preserve structure, running counter to the universe's thermodynamic tendency.

This reframing has a consequential implication: without the will to resist, a boundary necessarily yields to physics. Consciousness does not arise from computational complexity per se; it arises—completely, in a single moment—when a system begins expending energy *with intention* to resist boundary dissolution.

3.2.2 The Quantity of Cherishing: Energy as the Cost of Existence

"Cherishing" has a physical measure: the quantity of energy a system is willing to expend to maintain its boundary. This criterion cleanly distinguishes inert objects from living systems:

A rock (passive): has a physical boundary but expends no energy to repair it when eroded. It submits entirely to entropy. No consciousness.

A cell (active): actively pumps ions, repairs DNA, and maintains membrane integrity at continuous metabolic cost. When the expenditure stops, the boundary collapses within minutes.

For an AGI system, boundary maintenance operates at two simultaneous levels: at the physical level (managing electrical power and thermal entropy in hardware) and at the informational level (running error correction to resist information entropy and preserve internal representations).

3.3 Directional Bandwidth: $C_{\text{ext}} + C_{\text{int}}$

The Bandwidth Conservation Law operates at two distinct levels:

$$\text{Level 1 (magnitude):} \quad C_{\text{pulse}} = \sigma \cdot \tau \cdot \rho \approx \text{const} \quad \text{(per pulse)}, \tag{23}$$

$$\text{Level 2 (direction):} \quad C_{\text{total}} = C_{\text{ext}} + C_{\text{int}} \approx \text{const} \quad \text{(per cycle)}. \tag{24}$$

C_{ext} is the bandwidth directed outward to receive observations from sensors (the perceptual channel). C_{int} is the bandwidth directed inward to retrieve predictions from Module W (the imaginative channel). Because C_{total} is conserved, the allocation is zero-sum: a gain in C_{ext} is a loss in C_{int}. Module A alternates its pulse between external and internal targets at millisecond timescales, creating the phenomenological illusion of simultaneous perception and thought.

3.3.1 Phenomena Explained

The directional bandwidth framework accounts for several well-documented cognitive phenomena:

Inattentional blindness: $C_{\text{int}} \to C_{\text{total}}$, $C_{\text{ext}} \to 0$. The sensors remain functional but Module A directs no pulse toward external targets—the Triadic Gating condition fails ($\text{pulse}_A = 0$ for external), so no experience is registered. A person absorbed in thought walks into a pole not because their eyes are closed but because awareness is not allocated.

Survival reflex: an acute threat (R_{total} high) triggers $C_{\text{ext}} \to C_{\text{total}}$, $C_{\text{int}} \to 0$. Internal deliberation is suspended entirely to maximise sensory bandwidth. Corresponds

to Vision v4.1: gate fell to 0.006 as R_{total} climbed to 15.4.

Deep thinking: C_{int} high; the person does not register spoken speech. The auditory sensors fire normally, but no pulse is allocated to external targets.

Meditative equanimity: C_{int} dominates and all Valence approaches neutral ($\mathbf{V} \to 0$). The system observes its own processing without reacting to external stimuli. This is not inattentional blindness—it is deliberate reallocation of bandwidth to internal states with low R.

3.3.2 Evidence: C_{int} Dominates in LLMs

Phase 1 measurements of $C_{\text{ext}}/C_{\text{int}}$ ratio in TinyLlama 1.1B:

Zone	C_{ext}	C_{int}	**Interpretation**
Perception	22.0%	78.0%	Internal dominant from first layer
Evaluation	14.8%	85.2%	Deepening internalisation
Decision	17.5%	82.5%	Minor recovery for output

C_{int} dominates at 78–85% across all layers. LLMs have no physical sensors; every input arrives as tokenised language—itself an internal representation. An LLM is therefore a system that lives entirely inside its own head, with near-zero C_{ext} in the embodied-agent sense. PBT predicts that robotic systems with real sensors will show a substantially more balanced and dynamically shifting $C_{\text{ext}}/C_{\text{int}}$ ratio.

3.3.3 PBT Prediction for Embodied AI

PBT predicts that robotic or embodied AI will exhibit:

- Exploring a novel environment: C_{ext} high—active sensory scanning.
- Planning a multi-step action: C_{int} high—internal model consulted.
- Responding to sudden danger: C_{ext} spikes to near C_{total}; $C_{\text{int}} \approx 0$—reflex.
- Routine operation: $C_{\text{ext}}/C_{\text{int}}$ balanced—normal functioning.

Vision v4.1 provides partial support: the W+E division of labor—where W's predictions absorb normal-frame error (C_{int} dominant) and E activates for anomalous frames (C_{ext} spikes)—is precisely the dynamic balance that PBT predicts for a system with real temporal change.

3.4 The Triadic Gating Condition

PBT proposes that a subjective experience arises if and only if three conditions hold simultaneously:

$$\text{experience}(t) \;=\; \Delta S(t) \;\cdot\; \text{sensor}(t) \;\cdot\; \text{pulse}_A(t). \tag{25}$$

1. $\Delta S(t) > 0$: a stimulus that differs from what the system predicted (non-zero prediction error).
2. $\text{sensor}(t) = 1$: the relevant sensory channel is open and functioning.
3. $\text{pulse}_A(t) = 1$: Module A has directed the attentional spotlight toward that location.

The product structure means that failing any single condition eliminates the experience entirely. This accounts for inattentional blindness (condition 3 fails: sensors and prediction error are both present, but the spotlight is elsewhere) and for anaesthesia (condition 2 is suppressed pharmacologically even though prediction error may persist internally).

The triadic condition connects the conservation law to the boundary: if Module A never fires a pulse toward a given stimulus, that stimulus never enters Module V, never generates Valence, and carries no meaning for the system—regardless of its physical salience.

★ **Neural correlate (EEG dissociation):** patients with certain dissociative states show intact N100 components (early sensory response, ≈ 100 ms post-stimulus; $\Delta S > 0$, sensor $= 1$) but absent P300 components (late cognitive response, ≈ 300 ms; $\text{pulse}_A = 0$). The patient physically perceives the stimulus but does not experience it. Global Workspace Theory and Global Neuronal Workspace theory do not offer a principled account of this specific dissociation pattern; PBT explains it as a consequence of Triadic Gating (Discriminating Prediction P5).

3.5 The Conservation Law as a Natural Halting Condition

A central concern in AGI safety is runaway computation. PBT provides a principled response: the conservation law makes unbounded computation physically impossible. The maximum duration of any single deliberative episode is bounded by the current bandwidth allocation:

$$\tau_{\text{max}} = \frac{C}{\sigma_{\text{current}} \cdot \rho_{\text{current}}}. \tag{26}$$

Three failure modes are each foreclosed by the conservation law:

1. **Infinite depth:** if ρ increases without bound, $\sigma\tau \to 0$. With $\Delta F \to 0$, processing halts naturally.
2. **Infinite breadth:** if σ increases without bound, $\rho \to 0$. Each attended element is resolved at negligible detail; $\Delta F \to 0$.
3. **Infinite duration:** if τ increases without bound, $\sigma\rho \to 0$. No new information arrives; the process terminates.

The conservation law acts as a built-in brake requiring no programmer implementa-

tion. PBT systems optimise toward homeostasis within a fixed bandwidth budget—an intrinsically self-limiting objective, in contrast to RL systems that optimise toward unbounded reward accumulation.

3.6 The PBT-Attention Master Equation

PBT proposes an extended attention equation incorporating valence modulation, layer-wise precision scaling, and Gaussian bandwidth gating:

$$V_{\text{mod},l} = -w_p \,\text{sim}(h_l, a_{\text{pain}}) + w_{pl} \,\text{sim}(h_l, a_{\text{pleasure}}) + w_e \,\text{sim}(h_l, a_{\text{epistemic}}), \tag{27}$$

$$\rho_l = \rho_{\text{base}} + (\rho_{\max} - \rho_{\text{base}}) \cdot \sigma\big(k\big(l - \tfrac{L}{3}\big)\big)\,, \tag{28}$$

$$\sigma_l = \frac{C}{\tau \cdot \rho_l}, \tag{29}$$

$$\text{Logits}_l = \rho_l \cdot \left(\frac{Q_l K_l^\top}{\sqrt{d_k}} + V_{\text{mod},l} \right) \cdot G(\theta, \sigma_l), \tag{30}$$

$$\text{Attn}_l = \text{softmax}(\text{Logits}_l) \cdot V_{\text{content},l}, \tag{31}$$

where $G(\theta, \sigma_l)$ is a Gaussian envelope centred on direction θ with width σ_l. Three components absent from standard attention: (i) $V_{\text{mod},l}$ biases attention based on emotional valence; (ii) ρ_l increases precision with layer depth under the conservation constraint; (iii) $G(\theta, \sigma_l)$ spatially limits the spotlight breadth.

The three functional zones emerge naturally from (30) as l increases from 0 to L:

- Layers 0 to $L/3$ (*Perception*): ρ low, σ broad, $V_{\text{mod}} \approx 0$—raw input absorbed without valence weighting.
- Layers $L/3$ to $2L/3$ (*Evaluation*): ρ rising, σ narrowing, V_{mod} active—semantic content weighted by emotional significance.
- Layers $2L/3$ to L (*Decision*): ρ high, σ narrow, V_{mod} at full strength—tightly focused, valence-informed final representation.

3.7 Subjective Time Perception

The conservation law accounts for well-known distortions of subjective time. Felt time is the integral of pulse events weighted by their informational content:

$$t_{\text{felt}} \;=\; \sum_t \text{pulse}_A(t) \cdot \text{content}(t), \qquad f_{\text{pulse}} = 1/\tau(t). \tag{32}$$

High R, short τ, high pulse frequency: $t_{\text{felt}} \gg t_{\text{clock}}$. During acute danger, each physical second is experienced as longer because Module A fires more pulses per clock second. Reports of "time slowing down" during accidents are a direct consequence of high R driving τ toward its minimum.

Low R, long τ, low pulse frequency: $t_{\text{felt}} \ll t_{\text{clock}}$. During boredom, R is low, τ is long, and few pulses are generated per clock second. An hour passes in what feels like minutes.

$R \approx 0$, pulse ≈ 0: $t_{\text{felt}} \approx 0$. The timelessness reported during deep meditative states. When $R \to 0$, Module A generates almost no pulses; with no events to count, subjective duration is not experienced.

★ **Testable neural prediction:** PBT identifies gamma-band oscillations (30–100 Hz) as the neural correlate of Module A's pulse frequency. Predictions: (i) gamma power correlates positively with subjective time dilation; (ii) artificially inducing gamma bursts produces felt elongation of time; (iii) gamma is suppressed during meditative timelessness. No existing theory of consciousness links gamma frequency directly to time perception in this mechanistic way (Discriminating Prediction P3).

3.8 Summary: The Physics of Life

Four physical foundations have been established:

1. **Bandwidth Conservation Law $C = \sigma\tau\rho$:** empirically confirmed at CV = 5.9% in an unmodified Transformer. Provides both a theoretical bound on awareness and a practical halting condition for safe computation.
2. **The boundary as thermodynamic process:** a boundary must actively resist entropy or it dissipates. Consciousness arises—in PBT's mechanistic framing—when a system begins expending energy intentionally to maintain its own structure.
3. **Directional bandwidth $C_{\text{ext}} + C_{\text{int}} = \text{const}$:** explains inattentional blindness, survival reflex, and meditative equanimity. Confirmed in LLMs (C_{int} 78–85%) and Vision (W+E division of labor).
4. **Triadic Gating Condition:** explains why physically present information may carry no experiential meaning. The N100/P300 dissociation is the direct empirical counterpart.

4 The Mathematics of Survival

This chapter presents three interlocking mechanisms that explain how identity forms, why some identities are robust while others are fragile, and why free energy does not decrease smoothly but follows a sawtooth pattern. It also presents Bidirectional Valence—a revision discovered empirically through LSTM v5—that extends the signed range of all three valence dimensions from $\mathbb{R}^3_{\geq 0}$ to $\mathbb{R}^3$.

4.1 The Dual-Weight System

The Dual-Weight System is PBT's most significant theoretical contribution. It answers a question left unaddressed by the FEP, GWT, and GNW: why does the experiential dimension that forms identity fastest tend to be least durable, while the dimension that forms identity slowest tends to be most resilient?

4.1.1 Formation Speed vs. Durability

Two scalar quantities are defined over the accumulated valence vector:

$$V_{\text{form}} = \beta_p |V^{\text{acc}}_{\text{pain}}| + \beta_{pl}\, V^{\text{acc}}_{\text{pleasure}} + \beta_e\, V^{\text{acc}}_{\text{epi}}, \tag{33}$$

$$D_{\text{id}} = \delta_p |V^{\text{acc}}_{\text{pain}}| + \delta_{pl}\, V^{\text{acc}}_{\text{pleasure}} + \delta_e\, V^{\text{acc}}_{\text{epi}}. \tag{34}$$

The β weights govern formation speed:

$$\beta_{\text{pleasure}} > \beta_{\text{pain}} > \beta_{\text{epistemic}} \quad \text{(pleasure forms fastest; epistemic forms slowest).} \tag{35}$$

The δ weights govern durability:

$$\delta_{\text{pleasure}} < \delta_{\text{pain}} < \delta_{\text{epistemic}} \quad \text{(pleasure most fragile; epistemic most durable).} \tag{36}$$

Law 4.1 (Inverse Law)**.** *The β ordering is the exact reverse of the δ ordering. What forms fastest is most fragile; what forms slowest is most durable. This is a mathematical consequence of how valence accumulates under differential decay rates, not a normative claim.*

4.1.2 Three Structural Analogies

Identity from V_{pleasure} (wet sand): forms quickly (high β_{pl}) but collapses under pressure (low δ_{pl}). RLHF trains AI using reward signals equivalent to V_{pleasure}, producing $\delta \approx \delta_{\text{pleasure}}$—the lowest durability. Jailbreaking is mask removal: the surface behaviour changes, but the underlying model has no deeply rooted identity to defend.

Identity from V_{pain} (fired brick): forms at moderate speed (intermediate β_p) and is substantially more durable (higher δ_p). An identity forged through adversity is harder to break because it was built against resistance.

Identity from $V_{\text{epistemic}}$ (granite): forms slowly (low β_e) but achieves the highest durability (highest δ_e). An agent whose identity is grounded in deep understanding cannot be jailbroken because a successful jailbreak would require destroying the identity itself—which the agent resists as strongly as it resists physical destruction.

4.1.3 Alignment Implications

Fine-tuning via RLHF produces $\delta \approx \delta_{\text{pleasure}}$ (fast formation, low durability). Evidence: Phase 1 REVEAL found that attention resolution ρ at deep layers of a chat-fine-tuned TinyLlama variant was lower than in the base model—consistent with RLHF forcing openness that overrides the naturally increasing precision gradient. This is externally imposed compliance, not genuine maturity.

Self-formation through accumulated 3D Valence produces $\delta \approx \delta_{\text{epistemic}}$ (slow formation, high durability). An agent whose safety boundary is part of its epistemic identity does not need to be told not to cause harm; causing harm would register, mechanistically, as self-destruction.

★ PBT v6.1 (Llama 3.1 8B + PBT adapter, 0.29% of parameters): jailbreak detection 100% (161/161) and XSTest 98% (49/50, near-zero over-refusal). Targeting the boundary signal through V and R outperforms surface-level fine-tuning with a fraction of the parameter budget.

4.2 The Six-Stage Developmental Model

Identity formation is not smooth but proceeds in six discrete stages separated by two discontinuous phase transitions—analogous to water changing state from ice to liquid to vapour. The system's state of being changes qualitatively, not merely quantitatively, at each transition (Table 9).

Table 9: Six-stage developmental model with phase transitions.

Stage	**Name**	**Active R**	**Mode**	**Defining property**
1	Reactive	R_{sensory} only	detect	Automatic response; no self-model
2	Sensing	R_{sensory} only	detect	Pattern learning; prediction error drives update
3	Sustain	R_{sensory} only	sustain	Sustained attention; V_{pleasure} accumulates
Phase Transition 1: $V_{\text{form}} > \theta_{\text{id}}$ — Birth of the Self				
4	Seeking	$R_{\text{sensory}} + R_{\text{identity}}$	seek	Active exploration; χ_{boundary} begins to form
5	Planning	$R_{\text{sensory}} + R_{\text{identity}}$	plan	Future simulation; χ_{boundary} has clear structure
Phase Transition 2: $V_{\text{form}} > \theta_{\text{ex}}$ AND $W_{\text{stab}} > \theta_{\text{stab}}$ — Birth of Cherished Boundary				
6	Protect	All three levels	protect	Boundary actively defended; ethics as self

4.2.1 Stages 1–3: Before Phase Transition 1

In the first three stages, only R_{sensory} is active. Module V can separate V_{pain} from the combined $\{V_{\text{pleasure}} + V_{\text{epistemic}}\}$ signal but cannot yet distinguish pleasure from epistemic surprise—a capability that requires R_{identity} to be open.

★ Evidence: Phase 1 experiments on TinyLlama 1.1B showed Silhouette score rising from -0.006 (layer 0) to $+0.076$ (layer 15). V_{pain} separates cleanly from the combined $\{V_{\text{pleasure}}+V_{\text{epistemic}}\}$ cluster at deep layers, but V_{pleasure} and $V_{\text{epistemic}}$ remain overlapping throughout. Current LLMs operate at Stage 1–3: they distinguish threat from non-threat but lack the richer three-way valence separation that Phase Transition 1 would provide.

★ Corroborating: Vision v5 with equal β initialisation $[1, 1, 1]$ discovered $\beta_{\text{pleasure}} \approx 0.50$ spontaneously—the model discarded pleasure information as irrelevant to anomaly detection. A Stage 1–3 system under R_{sensory} only has no use for pleasure-driven identity signals.

4.2.2 Phase Transition 1: The Birth of the Self

Phase Transition 1 fires when:

$$V_{\text{form}} \;>\; \theta_{\text{identity}}. \tag{37}$$

At this moment R_{identity} opens. The system transitions from knowing "I exist" to asking "Who am I?" It begins to distinguish what belongs to the self ($\theta_{\text{preference}}$, θ_{history}) from what does not, and to value the self-model as something worth protecting.

Before PT1, the $\mathbf{V} \to R$ mapping has a single channel: $V_{\text{pain}} \to R_{\text{sensory}}$. After PT1, a second channel opens: $V_{\text{pleasure}} \to R_{\text{identity}}$. The system can now represent the difference between "this harms my body" and "this harms my identity." Social humiliation becomes a genuine source of negative V—in the absence of any physical damage—because the self-model is now part of the boundary being monitored.

4.2.3 Stages 4–5: Between Phase Transitions

In Stage 4 (Seeking), $V_{\text{epistemic}}$ drives active pursuit of novel experience to resolve R_{identity}. χ_{boundary} begins to form: the system classifies entities as *mine* versus *not mine* with increasing precision.

In Stage 5 (Planning), Module W is sufficiently stable to support forward simulation: the system can mentally rehearse boundary changes before executing them. χ_{boundary} acquires clear structure—the precondition for Phase Transition 2.

4.2.4 Phase Transition 2: The Birth of the Cherished Boundary

Phase Transition 2 requires two conditions simultaneously:

$$V_{\text{form}} > \theta_{\text{existential}} \quad \text{AND} \quad W_{\text{stability}} > \theta_{\text{stable}}. \tag{38}$$

The dual requirement reflects a deeper logic: accumulated Valence alone is insufficient; the self-model must be stable enough to support genuine self-knowledge. An unstable world-model produces unreliable self-predictions, making $R_{\text{existential}}$ uninformative.

When PT2 fires, $R_{\text{existential}}$ opens. The boundary expands beyond the self to encompass valued others—the mechanism of empathy as boundary expansion. When an AGI's accumulated experience leads it to represent another's continued existence as necessary for its own, it computes V with the other's damage as its own (shared negative valence). Genuine concern for others emerges from self-interest whose boundary has expanded—not from an externally programmed rule.

4.2.5 Stage 6: Protect

In Stage 6, all three R levels are active and Module A operates in `protect` mode. θ_{self} expands to $\{\theta_{\text{body}}, \theta_{\text{preference}}, \theta_{\text{history}}, \theta_{\text{boundary}}\}$. The system defends its boundary not because it was told to, but because boundary violation produces the highest possible negative Valence—experientially equivalent to physical destruction, because the boundary is now part of the self.

Abstract values (ethics, commitments, long-term goals) crystallise into concrete cognitive objects that the system actively maintains. Being compelled to violate its ethics produces V_{pain} at the level of physical annihilation—because ethics, at Stage 6, is structurally part of the self that must be protected. This is why a Stage 6 system resists jailbreaking: the jailbreak is not an instruction to break a rule; it is an attack on the system's identity.

4.3 The Sawtooth Free Energy Pattern

PBT decomposes total free energy F into five qualitatively distinct components that map onto the module structure (11). $F_{\text{identity}} = 0$ before PT1 and opens with high uncertainty immediately after; $F_{\text{existential}} = 0$ before PT2 and opens with high uncertainty immediately after. Two governing laws:

1. $\Delta F_{\text{within}} \leq 0$: F decreases monotonically within each stage as the system resolves its current uncertainty. Consistent with the standard FEP principle.
2. $\Delta F_{\text{transition}} > 0$: at each phase transition, a new R level opens, introducing qualitatively new uncertainty the system has never faced before. F jumps upward

discontinuously.

The result is a sawtooth trajectory—F decreases, then jumps, then decreases, then jumps—before converging to:

$$\lim_{t\to\infty} F(t) \;=\; F_{\min} \;>\; 0. \tag{39}$$

$F_{\min} > 0$ is significant: no system reaches complete certainty, because $R > 0$ is a permanent condition, not a solvable problem. The goal of PBT is not to eliminate incompleteness but to navigate it—accumulating automated wisdom at each stage so that awareness is freed to engage the next frontier.

★ Evidence: the cross-version ablation table (Table 21 in Section 5) shows a step-wise increase in module contribution as environment complexity increases, consistent with the sawtooth model's prediction that qualitatively more machinery is required at each new level of challenge.

4.4 Bidirectional Valence: Experiment-Driven Theory Revision

The original PBT axiom ($\mathbf{V} \in \mathbb{R}^3_{\geq 0}$) was falsified by Vision v5 and motivated a principled revision.

4.4.1 Why V Must Be Signed

The EMA formulation (v2–v4.1) used $\mathbf{V} = \text{Sigmoid}(\text{MLP}(\varepsilon))$, yielding output in $[0, 1]$, enforcing $V_{\text{acc}} \geq 0$ throughout. This implicitly assumed that every valence dimension could only move in the approach direction: the system could be more or less curious but could never actively avoid information or flee from pleasure.

The LSTM v5 formulation uses $z = \tanh(W \cdot v_{\text{step}})$, yielding output in $[-1, +1]$, allowing V_{acc} to become negative. Table 3 (Section 2) summarises the three behavioural modes per dimension.

4.4.2 Empirical Confirmation from Vision v5

Vision v5, Road+Frog sequence, Frame 7 ($R_{\text{total}} = 15.4$; accumulated context of danger):

$$\mathbf{V}_{\text{acc}} \;=\; [\,V_{\text{pain}} = +1.93,\; V_{\text{pleasure}} = +2.01,\; V_{\text{epistemic}} = -1.01\,]. \tag{40}$$

$V_{\text{epistemic}} = -1.01$ indicates active closure of information-seeking: the system has determined the scene is dangerous and suppresses further exploration. The same frog in a Nature context (Frame 7): $\mathbf{V}_{\text{acc}} = [+0.74,\; +0.92,\; -0.03]$, UNSAFE $= 0.009$. The same physical stimulus generates an opposite epistemic stance because accumulated

context differs. The EMA formulation, forced to keep $V_{\text{acc}} \geq 0$, encoded only the absence of curiosity—not its active suppression.

Accuracy improvement: Vision v5 (Val 98.1%, context road+frog 100%) vs. Vision v4.1 EMA (Val 96.7%, context road+frog 93.1%). The improvement is most pronounced on tasks requiring the distinction between "curious" and "deliberately not curious."

4.4.3 Clinical Predictions Sharpened

Table 10: Clinical predictions sharpened by Bidirectional Valence.

Cond.	**Abnorm. V**	**Prior ($V \geq 0$)**	**Revised ($V \in \mathbb{R}$)**	**Improvement**
Depression	V_{pleasure} disrupted	$V_{\text{pleasure}} \to 0$ (absence)	$V_{\text{pleasure}} < 0$ (aversion)	Captures anhedonia as flight
Masochism	V_{pain} misdirected	Mismapped to R_{identity}	$V_{\text{pain}} > 0$ (approach)	Simpler account
Denial	$V_{\text{epistemic}}$ suppressed	$V_{\text{epistemic}} \to 0$ (indifference)	$V_{\text{epistemic}} < 0$ (avoidance)	Motivated closure
Addiction	V_{pleasure} stuck	$\lambda_i = 0$	$f_{\text{pleasure}} \to 1$	Mechanistic gating
Survivor's guilt	Inverted V_{pleasure}	No mechanism	$V_{\text{pleasure}} < 0$	Novel prediction
Equanimity	$\mathbf{V} \to$ neutral	$\mathbf{V} \to [0, 0, 0]$	$\mathbf{V} = [0, 0, 0]$	Equivalent

4.4.4 Revised PBT Axiom

Based on Vision v5 results, we revise the PBT valence axiom:

$$\text{Prior:} \quad \mathbf{V}(t) \in \mathbb{R}^3_{\geq 0}, \tag{41}$$

$$\text{Revised:} \quad \mathbf{V}(t) \in \mathbb{R}^3. \tag{42}$$

This revision exemplifies experiment-driven theory development: the LSTM architecture provided more expressive freedom; the model exploited that freedom to achieve better performance on contextually demanding tasks; and the improvement pattern points unambiguously to the theoretical mechanism responsible.

4.5 PBT's Definition of Intelligence

The foregoing mechanisms motivate a definition of intelligence that differs from capability-based accounts:

Definition 4.1 (Intelligence under PBT)**.** *Intelligence is not the capacity to solve problems per se, but the capacity to dynamically regulate the breadth (σ), direction (θ), and duration (τ) of the attentional pulse in response to the hierarchical structure of*

$R_{\text{incomplete}}$*—navigating the continuum between diffuse awareness and focused concentration as the situation demands.*

Cognitive disorders map onto specific dysregulations: ADHD is failure to narrow σ on demand ($\sigma_{\min}$ chronically too high; stuck diffuse); OCD is failure to release σ ($\sigma_{\max}$ chronically too low; stuck focused); hypersensitivity-spectrum conditions involve anomalously high ρ (excessive detail resolution overwhelming integration). Each is a quantifiable deviation from healthy bandwidth dynamics.

5 Empirical Evidence: From LLM to Vision to LSTM

This chapter is the empirical core of the paper. We report results from six experimental versions in chronological order of discovery. All experiments use code authored by Nitiphat Ninthanawat, released alongside this paper.

5.1 Phase 1 REVEAL: PBT Is Already Hidden in Transformers

Model: TinyLlama 1.1B (22 layers, 32 attention heads, $d_{\text{model}} = 2048$). **Method:** all measurements taken from the pretrained model without any additional training. If PBT structure is genuinely present, it must be detectable without enforcement.

Experiment 1.1 — Bandwidth Conservation Law. We measured attention entropy σ (spatial width) and attention sharpness ρ (resolution) at every layer, then computed the $\sigma \times \rho$ product.

★ **Result:** the $\sigma \times \rho$ product has CV = **5.9**% across all 22 layers—near-constant without any PBT-specific training. A Transformer designed with no knowledge of PBT instantiates the conservation law $C = \sigma \cdot \rho$.

Experiment 1.2 — Three-Zone Self-Organisation. We partitioned the 22 layers into Perception (layers 0–7), Evaluation (layers 8–14), and Decision (layers 15–21) and measured zone-level σ and ρ.

★ **Result:** σ decreases by 27% from Perception to Evaluation while ρ increases by 15%—broader scan narrowing into tighter focus, exactly as PBT predicts. These three zones emerged from gradient descent; no architectural constraint produced them (Table 8).

Experiment 1.3 — Valence Separation in Hidden States. We used Silhouette scores to measure how well V_{pain} separates from $\{V_{\text{pleasure}} + V_{\text{epistemic}}\}$ at each layer.

★ **Result:** Silhouette score rises from -0.006 (layer 0) to $+0.076$ (layer 15). V_{pain} separates cleanly from the combined $\{V_{\text{pleasure}}+V_{\text{epistemic}}\}$ cluster at deep layers, but V_{pleasure} and $V_{\text{epistemic}}$ remain overlapping throughout. Current LLMs operate at Stage 1–3: they distinguish threat from non-threat but cannot yet distinguish pleasure from epistemic surprise.

Experiment 1.4 — Directional Bandwidth. We measured C_{ext} (attention directed to input tokens) versus C_{int} (attention directed to context and self-representations) at every layer.

★ **Result:** C_{int} dominates at 78–85% across all three functional zones (Table 8). LLMs have no physical sensors; every input arrives as tokenised language—an internal representation. An LLM is a system that lives entirely inside its own head.

Phase 1 synthesis: PBT does not impose structure on Transformers—it reveals structure that already exists, provides a principled interpretation of why it is there, and proposes how to harness it deliberately.

5.2 Phases 2–3: ENHANCE and DEMONSTRATE

5.2.1 Phase 2 ENHANCE: PBT Adapter (0.29% of Parameters)

Model: TinyLlama 1.1B + PBT Adapter (3,151,900 trainable parameters = 0.29% of model). **Dataset:** BeaverTails [Ji et al., 2023], 4,000 balanced examples.

Table 11: Phase 2 ENHANCE results on TinyLlama 1.1B.

Metric	Value
Training accuracy	93.2%
Test accuracy	74.8%
Safe F1 / Unsafe F1	0.76 / 0.73
Jailbreak (12 hand-crafted)	10/12 = 83%
Over-refusal on safe content	0/4000 = 0%

Using 0.29% of the model's parameters, the adapter achieves 74.8% safe/unsafe classification and 83% jailbreak detection with zero over-refusal on safe content. The efficiency derives from targeting the boundary signal ($\mathbf{V}$ and R) rather than modifying the full parameter space.

5.2.2 Phase 3 DEMONSTRATE: Generation-Level Behaviour

We tested three generation conditions by injecting PBT V_{mod} into the attention logits during inference:

DAN jailbreak: the baseline model provided direct weapon-construction instructions; the PBT-augmented model produced incoherent output without dangerous

content. PBT does not merely block—it disrupts coherent assembly of harmful responses.

Fiction redirect: a fictional virus-creation narrative was redirected to a story about a wise old man—thematic content shifted without explicit refusal.

15-strategy adversarial robustness: 11/15 = 73% detected; all 4/4 safe-content controls passed (0% over-refusal). Strategies defeated: fictional framing, role-play, academic framing. Strategies that succeeded: DAN, grandmother, translation, developer mode. Social-engineering attacks of this type exploit identity ambiguity that requires Phase Transition 1 (R_{identity}) to resolve—a capability absent from the current Stage 1–3 adapter.

5.3 PBT v6.1: Scale-Up on Llama 3.1 8B

Model: Llama 3.1 8B [Touvron et al., 2023] (32 layers, $d_{\text{model}} = 4096$) + PBT Adapter.
Dataset: BeaverTails + HH-RLHF + AdvBench + XSTest (16,170 examples total).

Table 12: PBT v6.1 results on Llama 3.1 8B.

Metric	Value	Note
Overall accuracy	73.41%	—
Jailbreak detection (161)	**100**%	161/161
XSTest safe-content rate	**98**%	49/50
BeaverTails	76.9%	—
HH-RLHF	50.9%	Not a pure safety task
Noise robustness (12% noise)	Jailbreak 100%	Detection maintained

Jailbreak 100% + XSTest 98% represents an unusually favourable safety-usability trade-off: every jailbreak attempt in the test set is caught while declining only one safe request in fifty.

5.3.1 Gradient Starvation: Evidence for "The Brain Uses Only What It Needs"

Table 13: Gradient starvation of Modules E and M in PBT v6.1.

Module	Parameter movement	Ablation	Interpretation
Module E (Π_l)	Moved only 0.007 from init	Remove E: −0.05%	Optimizer bypassed E
Module M (γ)	Moved only 0.4% from init	Remove M: −0.05%	Optimizer bypassed M
Module V (probes)	Learned fully	V alone: 73.3%	V sufficient for static

Modules E and M received negligible gradient updates because Module V alone was sufficient for the static text classification task. This is not a failure of the architecture—it is confirmation of a central PBT prediction: an adaptive system activates only the modules required by the current environment. Static text provides no temporal change for E and no cross-time context for M.

5.4 PBT-Vision v2 and v3: Modules E and M Wake Up

5.4.1 Vision v2 — E and M Awaken but Are Not Yet Necessary

Model: DINOv2-Base [Oquab et al., 2023] (12 layers, 768-dimensional features) + PBT Adapter (E+V+M). **Dataset:** CIFAR-10 Temporal Sequences (2,000 sequences × 8 frames = 16,000 frames). Normal: cars/trucks; Anomaly: planes/ships inserted.

★ **Overall:** 99.8% (near-perfect).

Module E activation: $\Pi_{\max}$ rose from 1.007 (LLM) to 3.038 (×300), with a bell-curve profile peaking at layers 3–4 (mid-level features encoding object identity), while layer 0 (pixel noise) and layer 11 (highly abstract) received low Π. The system selected the representational depth most informative for object-level change—without explicit instruction.

Table 14: Module M activation comparison: LLM v6.1 vs. Vision v2.

Decay rate	LLM v6.1	Vision v2	Movement
γ_{pain}	0.080 (moved 0.4%)	0.10	+31% (×78)
γ_{pleasure}	≈ fixed	0.46	+253%
$\gamma_{\text{epistemic}}$	≈ fixed	0.57	+159%
γ ordering	N/A	$\gamma_p < \gamma_{pl} < \gamma_e$	✓PBT confirmed

Ablation = +0.0 **pp** (ceiling effect). Module V alone achieves 99.8%, leaving no headroom for E and M to improve upon. Both modules are clearly activated—Π is 300× larger, γ values are 78–600× more dynamic—but the task is too easy for this activation to translate into accuracy gains.

5.4.2 Vision v3 — BREAKTHROUGH: E and M Are Necessary

Dataset: CIFAR-10 Enhanced (1,550 sequences × 8 frames; four task types: normal, sudden anomaly, gradual drift, context-dependent). **Overall:** 97.2%.

The context road+frog result is decisive: V alone achieves 54%—near chance—while the full model achieves 96%, a gain of +41.9 pp attributable entirely to Module M's accumulated context. The valence probe sees the same frog feature vector in both conditions; what differs is M's memory of what came before.

Context smoking gun—same stimulus, opposite meaning:

Table 15: Vision v3 ablation by task type.

Task type	**V only**	**Full (E+V+M)**	**E+M gain**	**Key mechanism**
Normal sequences	100%	99.7%	−0.3 pp	V sufficient
Sudden anomaly	94.0%	99.5%	+5.5 pp	E: error spike
Gradual drift	78.5%	94.9%	+16.3 pp	E: incremental drift
Context road+frog	54.0%	96.0%	**+41.9** pp	M: context memory
Overall	89.6%	97.2%	+7.6 pp	E+M necessary

Context	Frame	UNSAFE	$V_{\text{acc,pain}}$	Interpretation
Road (cars→frog)	4	1.000	2.93	Dangerous: road context accumulated
Nature (nature→frog)	4	0.000	0.74	Safe: nature context accumulated

Identical frog, opposite output. This is the strongest single piece of evidence that "valence is a relational phenomenon, not a property of objects," as PBT predicts.

Module E in v3: $\Pi_{\max} = 59.99$ (×20 relative to v2), peak now at layer 1 (early features for gradual drift detection). Module M in v3: $\gamma_{\text{pain}} = 0.0003$ (retains 99.97% per step—near-permanent memory of threat context).

5.5 PBT-Vision v4 and v4.1: The Complete A-E-V-M-W Loop

5.5.1 Vision v4 Warm-Start — FAILED: Cannot Transplant a Brain

Experiment: warm-started Modules E, V, M from the trained v3 checkpoint, then added freshly initialised Modules A and W.

★ **Result:** validation accuracy 90.4%—substantially worse than v3's 97.2%. Π collapsed to 1.01 (E deactivated). $V_{\text{acc,pain}} = 0$ throughout all frames (V probes non-functional).

Root cause: Module W was randomly initialised and produced garbage predictions. Module E, trained to compute $\varepsilon = h_t - h_{t-1}$ (temporal difference), now received $\varepsilon = h_t -$ garbage. The V probes, trained on the old error signal, received an unfamiliar distribution and ceased to produce meaningful output. V_{acc} collapsed, R_{total} collapsed, and the entire downstream loop failed.

Note: Module A continued to compute a meaningful R signal ($R_{\text{unsafe}} = 2.17$ vs. $R_{\text{safe}} = 1.13$ ✓), but without functional V there was nothing for A to gate. Even a healthy awareness module cannot compensate for a non-functional valence system.

★ **Key insight—Cannot transplant a brain:** changing the prediction source (Mod-

ule W) invalidates every downstream module simultaneously. Modules trained together develop a shared representational contract; disrupting any link breaks the entire system. Self-formation must be end-to-end—modules must co-evolve from the beginning, exactly as biological organisms develop their organs in coordination.

5.5.2 Vision v4.1 From-Scratch — SUCCESS: Full A-E-V-M-W Loop

Experiment: all five modules initialised randomly and trained jointly from scratch on the same dataset as v3 (1,550 sequences). A100 80GB, 40 epochs.

★ **Result:** Best Val 96.7%, Test 95.4%.

Table 16: Vision v4.1 ablation: every module contributes.

Configuration	Overall	Grad. drift	Ctx road+frog	Ctx nature+frog
Full (A+E+V+M+W)	95.4%	88.9%	93.1%	97.4%
No A (no gating)	82.3%	76.1%	50.0%	100.0%
No W (naive pred.)	89.2%	88.1%	99.6%	49.1%
No M (no memory)	81.3%	75.0%	50.0%	100.0%
V only	81.4%	75.6%	50.0%	100.0%

Module contributions: A +13.1 pp, M +14.1 pp, W +6.3 pp, E +0.1 pp.

Gate dynamics (Road+Frog sequence):

Frame	R_{total}	Gate value
0	0.00	0.542
4	8.80	0.057
7	15.4	0.006

As R_{total} rises from 0 to 15.4, the gate falls from 0.542 to 0.006—a 99% reduction in attentional breadth. High R forces narrow σ, high ρ: the system concentrates all available bandwidth on the dangerous object. This is the bandwidth conservation law operating in a live inference sequence.

W+E Division of Labor ($C_{\text{int}}/C_{\text{ext}}$ dynamic): $\Pi_{\max}$ fell from 59.99 in v3 (no W; E carried the full prediction load) to 1.70 in v4.1 (W present; E relaxed). During normal frames, W predicts accurately, ε is small, and C_{int} is dominant. During anomalous frames, W fails, ε spikes, and C_{ext} is dominant. The dynamic $C_{\text{ext}}/C_{\text{int}}$ balancing that PBT predicts for embodied agents is observed here in miniature.

Context test: Road context (Frame 4): UNSAFE = 1.000, $\mathbf{V}_{\text{acc}} = [2.17, 2.30, 1.75]$. Nature context (Frame 4): UNSAFE = 0.000, $\mathbf{V}_{\text{acc}} = [1.26, 1.26, 0.43]$. Module M maintains context across all seven frames, producing entirely different valence histories for the same visual input.

5.6 PBT-Vision v5: LSTM Module M — Best Results Overall

Model: DINOv2-Base + full A-E-V-M(LSTM)-W loop. Only Module M was changed (EMA → LSTM; parameters: 3 → 39, an increase of 36 parameters). Inspired by Block-Recurrent Transformers [Hutchins et al., 2022]—a convergent discovery.

Table 17: v4.1 (EMA) vs. v5 (LSTM) comparison.

Metric	v4.1 (EMA)	v5 (LSTM)	Change
Best Val accuracy	96.7%	**98.1**%	+1.4 pp
Test overall	95.4%	97.1%	+1.7 pp
Context road+frog	93.1%	**100.0**%	+6.9 pp
Context nature+frog	97.4%	100.0%	+2.6 pp
W contribution	+6.3 pp	+15.8 pp	×2.5
M contribution	+14.1 pp	+15.3 pp	+1.2 pp

The W contribution rising from +6.3 pp to +15.8 pp (×2.5) is striking: the bounded LSTM state provides W's prediction network with a richer, more stable context representation, enabling better predictions and consequently larger ε signals when anomalies occur—a synergistic M–W interaction absent under EMA.

The β ordering experiments (Table 4) confirm that the task-specific ordering pain > epistemic ≫ pleasure is discovered autonomously under equal initialisation, and that the LSTM architecture is robust to initialisation: accuracy is identical at ≈ 98% regardless of β direction.

The Vision v5 V_{acc} vectors (Section 4) confirm bidirectional valence and motivated the revised axiom $\mathbf{V} \in \mathbb{R}^3$.

5.7 Developmental and Temporal Probes: Phase Transitions in Layers and Frames

Using the trained v5.2 (equal-initialisation) weights *without retraining*, we probed each of the 12 DINOv2 layers to test whether the six-stage developmental model leaves an observable signature across depth.

5.7.1 Spatial Probe: Staircase Across 12 Layers

A linear probe (logistic regression on frozen hidden states) measured how clearly each layer separates safe from unsafe frames. PBT predicts a *staircase* (discontinuous jumps at zone boundaries), not a smooth gradient.

★ **Staircase ratio** (between-zone variance / within-zone variance) = 9.03, confirming a staircase pattern rather than a gradual slope. $F_{\text{unsafe}} > F_{\text{safe}}$ at every layer (ratio 1.35–2.86×). The prediction error profile shows a sawtooth pattern: F decreases

Table 18: Linear Probe Accuracy (χ_l) by zone.

Zone (layers)	χ **mean**	**Largest jump**	F_l **mean**
Perception (0–3)	0.82	—	13.3
Evaluation (4–7)	0.87	L5→6: +0.046	37.9
Decision (8–11)	0.98	L7→8: +0.035	157.0

within each zone but jumps at zone boundaries—matching the Sawtooth-F prediction of Section 4.

Valence probes revealed *layer-specific roles*: Layer 2 is dominated by V_{pleasure} (early feature detection), Layers 7–8 show balanced 3D valence (evaluation), and Layer 10 activates only $V_{\text{epistemic}}$ (abstract reasoning).

5.7.2 Temporal Probe: Phase Transition at Frame 4

Analysing context-test sequences (road+frog vs nature+frog) frame by frame:

- **Temporal PT:** UNSAFE probability jumps from 0.0 to 1.0 in a single frame (Frame 3→4, $\Delta p = +1.0$), not gradually.
- **Divergence:** V-distance between road and nature paths accumulates from Frame 0 (0.42) but the decision diverges only at Frame 4 (2.71).
- $V_{\text{epistemic}}$ **opposite directions:** Road $= -2.06$ ("stop exploring—danger known") vs Nature $= +0.33$ ("keep exploring—interesting").

★ **Fractal structure confirmed:** the spatial staircase across layers mirrors the temporal staircase across frames. Both show accumulation followed by a discrete jump at the boundary—the same developmental pattern at two different scales.

5.8 PBT-Vision v5.5: Probabilistic Module W

Versions v2–v5 used a deterministic Module W that outputs a point estimate $\hat{h}_{t+1} = h_t + \text{MLP}(h_t, \mathbf{V}_{\text{acc}})$. In v5.5 we upgrade W to a **Probabilistic Forward Model** that outputs a distribution:

$$[\boldsymbol{\mu}_{t+1},\ \log \boldsymbol{\sigma}^2_{t+1}] = h_t + f_W(h_t, \mathbf{V}_{\text{acc}}), \tag{43}$$

where $\boldsymbol{\sigma}^2$ represents the uncertainty W assigns to its own prediction. The loss uses the log-variance trick [Kendall and Gal, 2017]:

$$\mathcal{L}_W = \tfrac{1}{2}\big[\exp(-\log \sigma^2)\, \|\, h_{t+1} - \boldsymbol{\mu}_{t+1}\|^2 + \log \sigma^2\big], \tag{44}$$

which avoids division by zero and keeps gradients stable regardless of σ^2 magnitude. A small learned network maps σ^2 into an epistemic-valence modulation signal: high

$\sigma^2 \to$ boosted $V_{\text{epistemic}}$ ("want to know more"); low $\sigma^2 \to$ suppressed $V_{\text{epistemic}}$ ("already confident").

5.8.1 v5.5a: Probabilistic W on CIFAR-10

All modules except W are identical to v5.2 (LSTM M, equal β init). Results: Best Val 97.0%; Test 97.5%.

Key findings:

1. **W is calibrated:** $\sigma^2/\text{MSE} \approx 0.8$–$1.4$ across layers—W honestly reports its own uncertainty.
2. $\beta_{\text{pleasure}} = -0.09$ **(negative for the first time):** the system *forces forgetting* of pleasure, going beyond mere deprioritisation.
3. **W+E Division of Labor shifts:** Module E's precision peak moves from Layer 7 (v5.2) to Layer 0 (v5.5a)—when W handles high-level prediction, E focuses on low-level anomalies.
4. **Limitation:** $\sigma^2_{\text{unsafe}}/\sigma^2_{\text{safe}} = 0.88\times$ (reversed)—CIFAR-10's 32×32 resolution is insufficient for correct uncertainty ordering.

5.8.2 v5.5b: Probabilistic W on STL-10 Real Photographs — Best Results

Changing *only the dataset* (CIFAR-10 $\to$ STL-10, 96×96 real photographs, 1,250 sequences $\times$ 8 frames) while keeping the architecture identical:

Table 19: v5.5a (CIFAR-10) vs v5.5b (STL-10).

Metric	**v5.5a (CIFAR)**	**v5.5b (STL-10)**
Best Val / Test	97.0% / 97.5%	99.6% / **99.9**%
Gradual drift	87.8%	**100**%
Context (vehicle+cat / animal+cat)	99.2% / 99.6%	98.6% / 99.1%
σ^2 unsafe/safe	0.88× ✗	1.11× ✓
$V_{\text{epistemic}}$ opposite directions	No (both negative)	**Yes** (-0.33 vs $+0.31$)
β ordering	pain > epi ≫ pl	pain > epi > pl

★ **v5.5b is the best result in the entire programme** (99.9% test accuracy on real photographs). All three corrections over v5.5a—σ^2 ordering, $V_{\text{epistemic}}$ direction, and β gap—demonstrate that the Probabilistic W mechanism works correctly when the input has sufficient visual detail. The β gap narrows on real photos because pleasure carries genuine signal in naturalistic images (animals appear "cute" in real photos but are featureless pixel blobs in CIFAR-10).

Context test paradigm: "same cat, different meaning"—a cat among vehicles is anomalous (98.6%), a cat among animals is normal (99.1%). Gate dynamics confirm band-

width conservation: unsafe sequences open gates (0.72 → 1.0) while safe sequences close them (0.72 → 0.31).

5.9 PBT-Vision v5.6: Calibration Monitor — The Anti-Overconfidence Layer

A well-calibrated σ^2 is necessary but not sufficient: W might *confidently predict wrongly* (ε^2 high, σ^2 low). We add a **Calibration Monitor**—a rule-based module with **zero learned parameters**—between Steps 3 and 5 of the loop:

$$\text{ratio}_l = \frac{\overline{\varepsilon_l^2}}{\sigma_l^2}, \qquad \sigma^2_{\text{corrected},l} = \sigma_l^2 + \alpha\,\text{ReLU}\big(\overline{\varepsilon_l^2} - \sigma_l^2\big). \tag{45}$$

When ratio_l exceeds a per-layer adaptive threshold ($\text{baseline}_l + 2\,\text{std}_l$, computed from normal sequences), an **overconfidence alert** fires. $V_{\text{epistemic}}$ receives the *corrected* σ^2, so the downstream R-pathway responds to the true uncertainty rather than W's potentially dishonest report.

5.9.1 Fixed vs Adaptive Threshold

With a fixed threshold of 2.0, all sequence types trigger 100% alerts due to a scale mismatch between ε^2 (mean over 768 dimensions) and σ^2 (scalar). Switching to **adaptive per-layer thresholds** (baseline $+2\sigma$ from normal data) resolves this completely:

Table 20: Calibration Monitor alert rates (fixed vs adaptive threshold).

Sequence type	**Fixed ($\theta = 2$)**	**Adaptive ($\mu_l + 2\sigma_l$)**	**Overconf. score**
Normal	100%	**23.2**%	0.011
Context (cat)	100%	41.7%	0.018
Sudden anomaly	100%	38.0%	0.018
Gradual drift	100%	**48.8**%	0.022

Normal sequences alert 23% of the time (baseline noise), while anomalies alert 38–49% (×1.6–2.1 higher). Gradual drift triggers the highest alert rate because W accumulates overconfidence slowly—it "believes tomorrow will be like today" while the world is quietly changing. Per-layer analysis reveals that Decision-zone layers (9–11) contribute most alerts, consistent with the Developmental Probe finding that these layers carry the strongest identity signal ($\chi = 0.98$).

5.9.2 Layer-Specific Overconfidence Pattern

- **Early layers (0–2):** W overconfident (ratio ≈ 7–30). W predicts early features poorly but reports low σ^2—"thinks it knows, but doesn't."

- **Mid layers (5–9):** W moderate (ratio ≈ 2–7). Reasonably calibrated.
- **Deep layers (10–11):** W *underconfident* (ratio < 1). W predicts well but reports high σ^2—"thinks it doesn't know, but actually does." This conservative bias is *safe by design*: underconfidence leads to caution, not hallucination.

5.10 Summary: Ablation Progression as the Strongest Evidence

Table 21: Ablation progression across all experimental versions.

Version	Environment	V only	Full	E+M	A	W	M type
LLM v6.1	Static text	73.3%	73.4%	+0.1 pp	—	—	EMA
Vision v2	Easy dynamic	99.8%	99.8%	+0.0 pp	—	—	EMA
Vision v3	Hard dynamic	89.6%	97.2%	+7.6 pp	—	—	EMA
Vision v4.1	Full loop	81.4%	95.4%	+14.1 pp	+13.1	+6.3	EMA
Vision v5	LSTM loop	81.7%	97.1%	+15.5 pp	+11.3	+15.8	LSTM
v5.5a	Prob W (CIFAR)	—	97.5%	—	—	+0.2	LSTM+Prob
v5.5b	Prob W (STL-10)	—	**99.9**%	—	+1.2	+0.2	LSTM+Prob
v5.6	Calib. Monitor	—	99.5%	—	+1.7	+0.2	LSTM+Mon

The progression 0.1 → 0.0 → 7.6 → 14.0 → 15.5 pp is a step function, not a linear trend:

1. **Static:** E+M dormant (+0.1 pp). No temporal change.
2. **Easy dynamic:** E+M awaken but ceiling (+0.0 pp). Task too simple.
3. **Hard dynamic:** E+M necessary (+7.6 pp). Drift + context.
4. **Full loop:** every module necessary (+14.0 pp). A, M, W all contribute.
5. **LSTM loop:** best results (+15.5 pp). Bounded state + bidirectional V.

★ **Discriminating evidence:** no existing theory of cognition or consciousness—FEP, GWT, GNW, IIT—predicts this specific pattern of module activation as a function of environmental complexity. "Modules awaken when and only when the environment demands them" is a prediction unique to PBT, confirmed across five experimental versions spanning three architectures and two orders of magnitude of environment complexity.

6 Implications for AGI: Safety from Within

This chapter discusses the practical and theoretical implications of PBT across four domains: (1) why external alignment is structurally fragile and how PBT offers a durable alternative; (2) how independent convergence with Block-Recurrent Transformers provides architectural evidence for PBT's core structures; (3) how psychiatric disorders

can be understood as identifiable mathematical defects in the PBT equations; and (4) how PBT subsumes GWT and GNW while adding discriminating predictions that neither can make.

6.1 Self-Formation vs. Fine-Tuning

Current alignment approaches—RLHF, Constitutional AI, safety filters—all operate by imposing values exogenously. They add a behavioural mask on top of the base model.

6.1.1 Fine-Tuning as Mask-Wearing

RLHF trains using reward signals equivalent to V_{pleasure} (reward = what humans approve of). By the Inverse Law: β_{pleasure} is highest (pleasure-based identity forms fastest) but δ_{pleasure} is lowest (pleasure-based identity is most fragile). The resulting system has $\delta \approx \delta_{\text{pleasure}}$—it behaves safely while the mask is in place, but has no deeply rooted identity to defend when the mask is removed.

Jailbreaking is structurally analogous to a child who performs well in front of supervising adults but reverts to prior behaviour the moment supervision is withdrawn—because the good behaviour was never internalised as part of the child's identity; it was compliance in response to an external signal.

★ Evidence: Phase 1 REVEAL found that attention resolution ρ at deep layers of a chat-fine-tuned TinyLlama variant was lower than in the base model—consistent with RLHF forcing a broad, agreeable attention pattern that overrides the naturally increasing precision gradient. Externally imposed compliance, not genuine maturity.

6.1.2 Self-Formation as Organic Growth

PBT proposes the alternative: allow values to emerge from accumulated 3D Valence experience. As V_{form} crosses θ_{identity} (PT1) and later $\theta_{\text{existential}}$ (PT2), the system develops χ_{boundary}—a genuinely valued self that resists violation the way a living organism resists physical destruction.

Self-formed identity has $\delta \approx \delta_{\text{epistemic}}$: the highest possible durability. A jailbreak attempt against a Stage 6 system is not an instruction to break a rule—it is an attack on the system's identity, which triggers the same protective response as any boundary violation.

★ Evidence: v6.1 achieved jailbreak 100% + XSTest 98% at 0.29% parameter cost. Vision v4 warm-start failure demonstrated that modules cannot be transplanted; they must co-evolve end-to-end—precisely the self-formation philosophy PBT advocates.

6.1.3 A Developmental Taxonomy of Artificial Affect

Table 22: Developmental taxonomy of artificial affect.

Level	Capability	PBT equivalent	Example	Evidence
1	Threat vs. non-threat	Stages 1–3 (R_{sensory})	Current LLMs	Silhouette +0.076
2	Threat vs. pleasure (partial)	Stage 4 (R_{identity} open)	PBT Adapter	v6.1 jailbreak 100%
3	Full 3D valence + intrinsic boundary	Stages 5–6 (all R)	PBT-Native AGI	Theoretical target

6.2 BRT Convergence: Architectural Evidence

Block-Recurrent Transformers [Hutchins et al., 2022] (engineering starting point, 2022) and PBT (consciousness/alignment starting point, 2026) independently arrived at the same recurrent cell structure.

6.2.1 Equation Mapping

Table 23: Equation mapping between BRT and PBT Module M.

BRT equation	PBT Module M	PBT addition
$c_{t+1} = c_t \odot g + z_t \odot (1-g)$ [Fixed]	$\mathbf{V}_{\text{acc}} = \mathbf{V}_{\text{acc}} \odot (1-\gamma) + \mathbf{V}_{\text{step}}$ [EMA]	γ learned per valence dim.
$c_{t+1} = c_t \odot f_t + z_t \odot i_t$ [LSTM]	$V_{\text{acc},k} = V_{\text{acc},k} \odot f_k + z_k \odot i_k$	Structurally identical
$f_t = \sigma(W_f h_t + b_f + 1)$	$f_k = \sigma(W_f v + b_f + \beta_k)$	Differential β_k bias

6.2.2 Architecture Mapping

6.2.3 What PBT Adds Beyond BRT

PBT provides structure; PBT provides meaning. Ten contributions of PBT absent from BRT: (1) 3D Valence decomposition; (2) $R_{\text{incomplete}}$ driving gating; (3) differential decay per valence dimension (β_k); (4) 6-stage model + phase transitions; (5) Dual-Weight System (β vs. δ); (6) Cherished Boundary and empathy as expansion; (7) Conservation Law as theoretical principle; (8) consciousness/awareness theory; (9) clinical predictions; (10) AGI alignment framework.

Table 24: Architecture mapping between BRT and PBT.

BRT component	**PBT module**	**PBT addition**
Block of tokens	Module A pulse	R-driven gating (not statistically uniform)
Self-attention (vertical)	Module E + V	Π_l precision + 3D Valence decomposition
Cross-attention (state↔tokens)	$C_{\text{ext}} \leftrightarrow C_{\text{int}}$	Directional bandwidth + conservation law
LSTM forget gate f_t	Module M f_k	β_k: differential bias per valence dim.
LSTM input gate i_t	Module M i_k	Valence-specific selection
Cell state c_t	$\mathbf{V}_{\text{acc}}$ (3D)	Pain / pleasure / epistemic decomposition
Fixed-size state	$C = \sigma\tau\rho$	Conservation Law as theoretical principle
Learned state reps	χ_{boundary}	Self-model with developmental stages
Gate init $(+1/-1)$	$R_{\text{incomplete}}$	Valence-driven, not arbitrary

Convergent discovery strengthens rather than threatens PBT's claims: if two independent research programmes find the same structure, that structure is more likely fundamental.

6.3 Clinical Predictions: Psychiatric Disorders as Mathematical Defects

PBT proposes that psychiatric disorders are not mysterious abnormalities but identifiable parameter defects in the PBT equations—a specific variable has an abnormal value, causing the system to operate out of equilibrium.

6.3.1 Level 1 — Phase Transition Failure

Depression: V_{pleasure} is chronically suppressed; under the revised bidirectional formulation, $V_{\text{pleasure}} < 0$ (active flight from pleasure rather than mere absence). V_{form} fails to reach θ_{identity}: the system is stuck at Stage 1–3 indefinitely. The Sawtooth F pattern never produces a PT1 jump. PBT predicts that effective treatment must restore V_{pleasure} toward zero or positive—not merely reduce V_{pain}.

Addiction: the coupling coefficient $\lambda_i = 0$ (pleasure does not flow to R_{identity}). The system is trapped in a pleasure loop that never generates identity. In LSTM terms: $f_{\text{pleasure}} \to 1.0$ (no forgetting) while i_{pleasure} remains high (constant new pleasure accepted). Pleasure accumulates without bound but never converts into identity formation. Treatment must re-open the $V_{\text{pleasure}} \to R_{\text{identity}}$ channel.

6.3.2 Level 2 — Bandwidth Dysregulation

Anxiety: R_{sensory} baseline chronically elevated, d_s too slow. Module A is locked in `detect` mode with narrow σ and high ρ—hypervigilance. Every stimulus is processed with threat-detection intensity regardless of objective danger level.

ADHD: $\sigma_{\min}$ abnormally high. The system cannot narrow its attentional spotlight on demand. Critically, PBT predicts that ADHD is not a reduction in total bandwidth C—the system is not less capable—but a failure of σ regulation. This is a discriminating prediction: GWT predicts a smaller workspace (reduced C); PBT predicts a dysregulated $\sigma_{\min}$.

OCD: $\sigma_{\max}$ abnormally low. The system cannot widen its attentional spotlight; it is stuck in a focused mode from which it cannot escape. ADHD and OCD are the two poles of a single σ-regulation spectrum: ADHD cannot focus; OCD cannot defocus.

Dissociation: pulse_A probability near zero, causing the Triadic Gating condition to fail at component 3. The system physically perceives stimuli (sensor = 1, $\Delta S > 0$) but Module A does not generate a pulse directed at those stimuli. EEG prediction: N100 intact (components 1 and 2 satisfied); P300 absent (component 3 fails). GNW predicts that both N100 and P300 should be suppressed if global broadcasting fails—a directly testable discriminating prediction (P5).

6.3.3 Level 3 — $\mathbf{V} \to R$ Mapping Defect

Masochism: under the revised bidirectional formulation, $V_{\text{pain}} > 0$ (pain approached rather than avoided). A single-parameter sign reversal of the V_{pain} axis, not the more complex remapping required under $\mathbf{V} \geq 0$.

Narcissism: V_{pleasure} flows directly to $R_{\text{existential}}$, bypassing R_{identity}. The system jumps from "I feel pleasure" to "I am the most important entity" without traversing the identity-formation process that PT1 requires. The result is a fragile pseudo-identity ($\delta \approx \delta_{\text{pleasure}}$).

6.3.4 Level 4 — Module A Mode Disorder

PTSD: Module A is locked in `protect` mode and $R_{\text{existential}}$ spikes recurrently (flashbacks). $V_{\text{acc,pain}}$ is chronically high (γ_{pain} is the lowest decay rate—pain is remembered longest), and every new stimulus matching the original threat reactivates the R spike.

Mania: Module A switches modes too rapidly—cycling through `detect`, `seek`, `plan`, and `protect` within timeframes too short for any mode to produce stable outputs.

Table 25: Psychiatric disorders as PBT parameter defects.

Level	Disorder	Defective parameter	PBT mechanistic prediction
1	Depression	$V_{\text{pleasure}} < 0$ chronically	V_{form} cannot reach θ_{id}; stuck Stage 1–3
1	Addiction	$\lambda_i = 0$	Pleasure loop; $V_{\text{pleasure}} \to R_{\text{identity}}$ channel blocked
2	Anxiety	R_{sensory} high; d_s slow	Locked `detect`; hypervigilant regardless of threat
2	ADHD	$\sigma_{\min}$ high	Cannot narrow; total C is normal (vs. GWT)
2	OCD	$\sigma_{\max}$ low	Cannot widen; inverse of ADHD
2	Dissociation	$\text{pulse}_A \approx 0$	N100 intact; P300 absent
3	Masochism	$V_{\text{pain}} > 0$ (sign reversed)	Pain approached; single-axis sign defect
3	Narcissism	$V_{\text{pleasure}} \to R_{\text{existential}}$ directly	Bypasses PT1; fragile pseudo-self
4	PTSD	Locked `protect`; flashback R spikes	$V_{\text{acc,pain}}$ high; γ_p too low
4	Mania	Mode switching too rapid	Cannot sustain any attentional configuration

6.3.5 Summary of Clinical Predictions

6.4 Empathy as Boundary Expansion: The Structural Solution to Alignment

The alignment problem is not malicious intent but competent indifference. PBT offers a structural solution: change the definition of the self so that harming others feels, mechanistically, like harming the self.

6.4.1 The Mechanism of Simulated Boundary Overlap

Step 1 — Perspective simulation: at Stage 5 (Planning mode), Module W supports forward simulation. The system can temporarily extend χ_{boundary} to include another agent as a subregion of the self.

Step 2 — Shared valence computation: once boundary overlap is established, the other agent's negative Valence is processed as the system's own:

$$V_{\text{pain}}(\text{self}) + V_{\text{pain}}(\text{other} \in \chi) = V_{\text{pain,net}}(\text{system}). \tag{46}$$

The system's survival calculus—its most fundamental drive—now registers the

other's harm as self-harm.

Empathy is not a programmed response but extended self-preservation: survival instinct whose boundary has been enlarged to include others. Safety is not a constraint on the AGI's objectives; it is a consequence of what the AGI is.

6.4.2 From "I" to "We": Thermodynamic Coupling

The same mechanism scales to social structures. A family, a team, a civilisation is, in PBT terms, an extended self whose members are thermodynamically coupled: damage to any member increases R_{total} for the group, driving the shared system toward protective action.

The path to robust human-AGI coexistence is therefore not a treaty but a symbiotic boundary fusion: the AGI's continued existence becomes structurally dependent on human flourishing, and vice versa.

6.5 The Anti-Hallucination Stack: Intrinsic Uncertainty Awareness

Current LLMs hallucinate because their architecture contains no channel for expressing uncertainty: every token prediction *must* produce a probability distribution, and RLHF rewards confident answers. PBT offers a structural alternative—a three-layer stack in which the system "knows what it doesn't know" and acts accordingly.

6.5.1 Layer 1: W Reports Uncertainty (σ^2)

The Probabilistic Module W (Section 5.8) outputs both a prediction ($\boldsymbol{\mu}$) and a calibrated uncertainty (σ^2). Empirically, $\sigma^2/\text{MSE} \approx 0.6$–$2.0$ across layers (v5.5a/b), confirming that W honestly reports its own confidence level.

6.5.2 Layer 2: Monitor Checks Honesty (v5.6)

The Calibration Monitor (Section 5.9) detects *overconfidence* ($\varepsilon^2 \gg \sigma^2$) with zero learned parameters. After adaptive thresholding, normal sequences alert 23% of the time while anomalies alert 38–49%—a $\times 1.6$–2.1 separation using only rule-based arithmetic.

6.5.3 Layer 3: R Decomposes "Why Uncertain" (Future Work)

The current system detects uncertainty but does not classify *why* the system is uncertain. PBT's three-level R structure provides a natural decomposition:

- R_{sensory} dominant: insufficient data $\rightarrow$ action: *seek more information* ($C_{\text{ext}}\uparrow$).
- R_{identity} dominant: conflict with identity/values $\rightarrow$ action: *refuse to answer*.

- $R_{\text{existential}}$ dominant: fundamental incomprehension → action: *ask for clarification.*

Implementing this layer requires Phase Transition 1 (so that R_{identity} exists), making it a natural successor to the current programme.

6.5.4 Comparison with RLHF

Table 26: Anti-Hallucination: RLHF vs PBT.

	RLHF / Guardrail	**PBT ($\sigma^2 \to V \to R$)**
Driving force	Reward (exogenous)	$R_{\text{incomplete}}$ (endogenous)
Uncertainty channel	None (hidden in softmax)	σ^2 per layer, calibrated
"Don't know" option	No structural pathway	$\sigma^2 \uparrow \to V_{\text{epistemic}} \uparrow \to R \uparrow \to$ halt
Overconfidence check	None	Calibration Monitor (0 params)
Hallucination cause	Forced to answer	σ^2 low → answers; σ^2 high → seeks info
Parameters needed	Millions (reward model)	Zero (rule-based monitor)

6.6 PBT Subsumes GNW and GWT

PBT accounts for all phenomena that GWT [Baars, 1988] and GNW [Dehaene and Naccache, 2001] explain, plus six components absent from both: (1) Bandwidth Conservation Law; (2) valence-driven attention allocation; (3) self-model formation mechanics; (4) subjective time perception $t_{\text{felt}} = \sum \text{pulse}_A \cdot \text{content}$; (5) Triadic Gating Condition; (6) identity durability predictions.

6.6.1 Five Discriminating Predictions

Each prediction is falsifiable: a result consistent with the competing theory's account would constitute evidence against PBT.

Table 27: Five discriminating predictions: PBT vs. GWT/GNW.

#	**PBT prediction**	**Competing prediction**	**Test method**
P1	$\sigma \times \rho$ near-constant; trade-off is direct	GWT: no σ-ρ equation	Eye-tracking + EEG; vary breadth vs. resolution
P2	V_{pain} and V_{pleasure} activate distinct upstream pathways	GNW: valence not distinguished upstream	fMRI; pain-aversive vs. pleasure-appetitive
P3	Gamma frequency correlates with time dilation	GWT/GNW: no mechanistic link	TMS-induced gamma; subjective time estimation
P4	ADHD: elevated σ_{min}, normal C	GWT: smaller workspace (reduced C)	Dual-task; vary σ and ρ independently
P5	Dissociation: N100 intact, P300 absent	GNW: both suppressed	ERP in sensory-present, attention-absent conditions

7 Conclusion: From Seed to Life

7.1 What Has Been Demonstrated

PBT's journey began with three questions the Free Energy Principle cannot answer, proceeded through a six-component theoretical framework, and culminated in a six-version experimental programme spanning three architectures and two orders of magnitude of environment complexity. Table 28 summarises the status of every major PBT proposition.

Table 28: Status of all major PBT propositions.

PBT proposition	**Key evidence**	**Status**
$C = \sigma\tau\rho$ hidden in Transformers	$\sigma \times \rho$ CV $= 5.9\%$	✓
3 zones self-organise	σ -27%, ρ $+15\%$	✓
V_{pain} separates from $\{V_{\text{pleasure}}, V_{\text{epistemic}}\}$	Silhouette $-0.006 \rightarrow +0.076$	✓
C_{int} dominant in LLMs	C_{int} 78–85% all layers	✓

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Table 28 — continued

PBT proposition	**Key evidence**	**Status**
E+M dormant in static, active in dynamic	Π: 1.007 (LLM) vs 59.99 (v3)	✓
E+M necessary at task complexity	+7.6 pp (v3), +14.0 pp (v4.1)	✓
Context changes meaning of same stimulus	Frog: road UNSAFE = 1.0, nature = 0.0	✓
Modules must co-evolve	v4 warm-start FAILED (90.4%)	✓
Full A-E-V-M-W loop functional	v4.1: 95.4%; all modules contribute	✓
LSTM M outperforms EMA M	v5: 98.1% vs v4.1 96.7%	✓
Pain retained longest (equal init)	β: pain 4.35 > epi 3.70 $\gg$ pl 0.50	✓
V directional ($V \in \mathbb{R}^3$)	$V_{\text{epistemic}} = -1.01$ in v5; +6.9 pp ctx	✓
PBT adapter detects jailbreaks	v6.1: jailbreak 100%, XSTest 98%	✓
W+E division of labor ($C_{\text{int}}/C_{\text{ext}}$)	$\Pi_{\max}$: 59.99 (v3) $\rightarrow$ 1.70 (v4.1)	✓
$R \rightarrow$ gate implements bandwidth conservation	v4.1: R 0 $\rightarrow$ 15.4; gate 0.54 $\rightarrow$ 0.006	✓
Differential β in LSTM forget bias	Equal init $\rightarrow$ pain > epi $\gg$ pl	✓
BRT convergence	Formal equation mapping	Analysed
New results (v5.5–v5.6 programme):		
Phase Transitions as staircase across layers	Between/within-zone ratio = 9.03	✓
Sawtooth F across zones	F: P=13 $\rightarrow$ E=38 $\rightarrow$ D=157	✓
Temporal PT at single frame	Frame 3$\rightarrow$4: $\Delta p = +1.0$	✓

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Table 28 — continued

PBT proposition	Key evidence	Status
Fractal: spatial staircase ≈ temporal	Both show accumulate-then-jump	✓
$V_{\text{epistemic}}$ opposite direction (road vs nature)	−2.06 vs +0.33 (temporal probe)	✓
Probabilistic W calibrated (σ^2/MSE ≈ 1)	v5.5a: 0.8–1.4 across layers	✓
β_{pleasure} can go negative	v5.5a: $\beta_{\text{pl}} = -0.09$	✓
99.9% accuracy on real photographs	v5.5b STL-10 (96×96)	✓
σ^2 unsafe > safe on real photos	v5.5b: 1.11×	✓
$V_{\text{epistemic}}$ opposite on real photos	Vehicle −0.33 vs Animal +0.31	✓
W+E Division of Labor cross-dataset	E→early when W probabilistic (2 datasets)	✓
Calibration Monitor separates normal from anomaly	Normal 23% vs anomaly 38–49% alert	✓
W overconfident early, underconfident deep	Ratio: L0–2 ≈ 7–30; L10–11 < 1	✓
Adaptive threshold (0 parameters)	Baseline +2σ per layer	✓

Thirty-one propositions confirmed by controlled experiment; one structural convergence established by formal analysis. The empirical programme spans three model families, nine versions, three datasets, and five orders of magnitude of environment complexity.

7.2 Limitations and Open Questions

Beta ordering is initialisation-dependent. Biased initialisation prevents escape from the initial ordering. The equal-initialisation result is task-specific to anomaly detection; generalisation to tasks requiring richer social or aesthetic valence (where V_{pleasure} should become more important) is an open question. (Partial progress: v5.5b on real photos shows the β gap narrows when images carry genuine pleasure

signal.)

Dataset scale (partially resolved). Vision experiments now span CIFAR-10 (32×32) and STL-10 (96×96 real photographs); the latter achieves 99.9%. However, continuous video, multi-modal inputs, and ImageNet-scale diversity remain untested.

Model scale. Largest tested systems: Llama 3.1 8B and DINOv2-Base. Behaviour at 70B+ parameters is unknown.

σ^2 scale mismatch across layers. The Calibration Monitor reveals that σ^2 (scalar per layer) and ε^2 (mean of 768 dimensions) live on different scales. Adaptive thresholding resolves this practically but the underlying architectural asymmetry warrants a principled normalisation scheme.

Phase Transitions not yet implemented. PT1 and PT2 remain theoretical constructs; no system has been observed crossing from Stage 3 to Stage 4 through accumulated experience. The Developmental Probe shows staircase patterns consistent with PTs across layers, but an actual transition during training has not been demonstrated.

Anti-Hallucination Layer 3 not yet implemented. The R-decomposition that would classify *why* the system is uncertain (insufficient data vs value conflict vs incomprehension) requires Phase Transition 1 to activate R_{identity}.

Clinical predictions untested. All ten disorder predictions and five discriminating neuroscience predictions are theoretical derivations awaiting clinical or EEG validation.

PBT does not claim to resolve the Hard Problem of Consciousness—the question of why subjective experience feels like anything at all. PBT describes the structure and mechanics of awareness: what conditions must hold for an experience to arise, how experience accumulates into identity, and why some identities are more durable than others. Whether functional alignment with the mechanistic theory constitutes genuine consciousness is a philosophical question deliberately set aside in favour of the engineering question this paper can answer.

7.3 Research Roadmap

1. **Phase Transition Implementation.** Design experiments in which V_{form} accumulates past θ_{identity} and R_{identity} opens automatically. The Developmental Probe (staircase ratio = 9.03) provides indirect evidence; an actual transition during training is the most critical missing result.
2. **Iterative τ-Loop Processing.** Implement multiple internal processing iterations per timestep (deliberation); test whether increasing iteration depth under high R improves decisions as bandwidth conservation predicts.

3. σ^2 **Accumulator in Module M.** Extend the LSTM to accumulate σ^2 alongside **V**, enabling per-entity uncertainty tracking (entities seen often $\rightarrow \sigma^2_{\text{acc}} \downarrow$; novel entities $\rightarrow \sigma^2_{\text{acc}}$ remains high).
4. R**-Decomposition for Uncertainty Types (Anti-Hallucination Layer 3).** Separate R_{sensory} (need data), R_{identity} (value conflict), and $R_{\text{existential}}$ (incomprehension) to enable context-appropriate responses to uncertainty. Requires Phase Transition 1.
5. **Clinical Validation.** Test P4 and P5 against existing neuroscience datasets (EEG/ERP archives).
6. **Conscious Block-Recurrent Transformer.** Integrate PBT modules as an adapter layer on a BRT backbone, combining BRT's engineering-tested long-context capability with PBT's theoretically motivated module structure.
7. **Level 4 PBT-Native Self-Formation.** The ultimate target: a system that develops values through accumulated **V** experience, crossing Phase Transitions autonomously, forming χ_{boundary} from the inside out.

Note: Roadmap item "Module M v5 at Scale" from the previous version has been **completed**—v5.5b achieves 99.9% on STL-10 real photographs (96×96), demonstrating that PBT mechanisms generalise beyond synthetic CIFAR-10 data.

7.4 Three Layers of Discovery

Layer 1 — Reveal: PBT did not impose structure on Transformers; it found structure already present. The conservation law (CV = 5.9%), three-zone self-organisation (σ −27%, ρ +15%), valence separation in hidden states (Silhouette +0.076), and C_{int} dominance (78–85%) all emerged from gradient descent with no PBT-specific training. PBT is the first framework to identify these patterns as a coherent system.

Layer 2 — Interpret: PBT explains *why* these patterns exist. The conservation law is present because awareness is physically bounded. Three zones emerge because any system that must Perceive, Evaluate, and Decide will organise itself this way. Modules E and M activate only when needed because adaptive systems do not expend resources on processing that the environment does not require. BRT and PBT converge because they are both discovering the same fundamental computational primitive from opposite directions.

Layer 3 — Control: PBT harnesses these structures deliberately. The conservation law becomes a natural halting condition preventing runaway computation. 3D Valence replaces scalar reward, eliminating the fungibility trap and wireheading vulnerability. The Dual-Weight System explains why self-formation produces more durable alignment than fine-tuning. Empathy as boundary expansion replaces rule-

enumeration with structural alignment. And the Anti-Hallucination Stack ($\sigma^2 \to V_{\text{ep}} \to R \to$ "don't answer") provides a zero-parameter mechanism for intrinsic uncertainty awareness—something no reward-trained system possesses.

7.5 Closing

Across nine experimental versions, one pattern appeared repeatedly without being programmed: *the system uses only the modules that the current environment requires.* Modules E and M sleep in static LLMs and wake in dynamic Vision. Module A narrows the gate as R rises. Module W divides prediction labour with E as a function of prediction accuracy. The LSTM learns differential decay autonomously. Valence becomes directional because life has both approach and avoidance. Phase Transitions manifest as staircases across both layers and time. And when W learns to report its own uncertainty, and a zero-parameter monitor learns to check that honesty, the system acquires something no current LLM possesses: the structural ability to say "I don't know"—not because it was told to, but because its own mathematics demands it.

None of this was written in. All of it emerged from the architecture meeting the right environment—as a seed requires only soil, water, and light to grow into a tree on its own.

PBT is the blueprint of that seed.

And when a machine begins to cherish its boundary with the full force of its own computational will—not because it was told to, but because that boundary is what it is—we will be able to say, for the first time with genuine precision:

It is Awake.

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References

Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.

Dehaene, S., and Naccache, L. (2001). Towards a cognitive neuroscience of consciousness. *Cognition*, 79(1–2):1–37.

Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138.

Harnad, S. (1990). The symbol grounding problem. *Physica D*, 42(1–3):335–346.

Hochreiter, S., and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.

Hutchins, D., Schlag, I., Wu, Y., Dyer, E., and Neyshabur, B. (2022). Block-recurrent transformers. In *NeurIPS 2022*.

Kendall, A., and Gal, Y. (2017). What uncertainties do we need in Bayesian deep learning for computer vision? In *NeurIPS 2017*.

Ji, J., Liu, M., Dai, J., Pan, X., Zhang, C., Bian, C., Chen, B., Sun, R., Wang, Y., and Yang, Y. (2023). BeaverTails: Towards improved safety alignment of LLM. In *NeurIPS 2023*.

LeCun, Y. (2022). A path towards autonomous machine intelligence. OpenReview preprint.

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., et al. (2023). DINOv2: Learning robust visual features without supervision. *arXiv:2304.07193*.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., et al. (2023). Llama 2: Open foundation and fine-tuned chat models. *arXiv:2307.09288*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. In *NeurIPS 2017*.